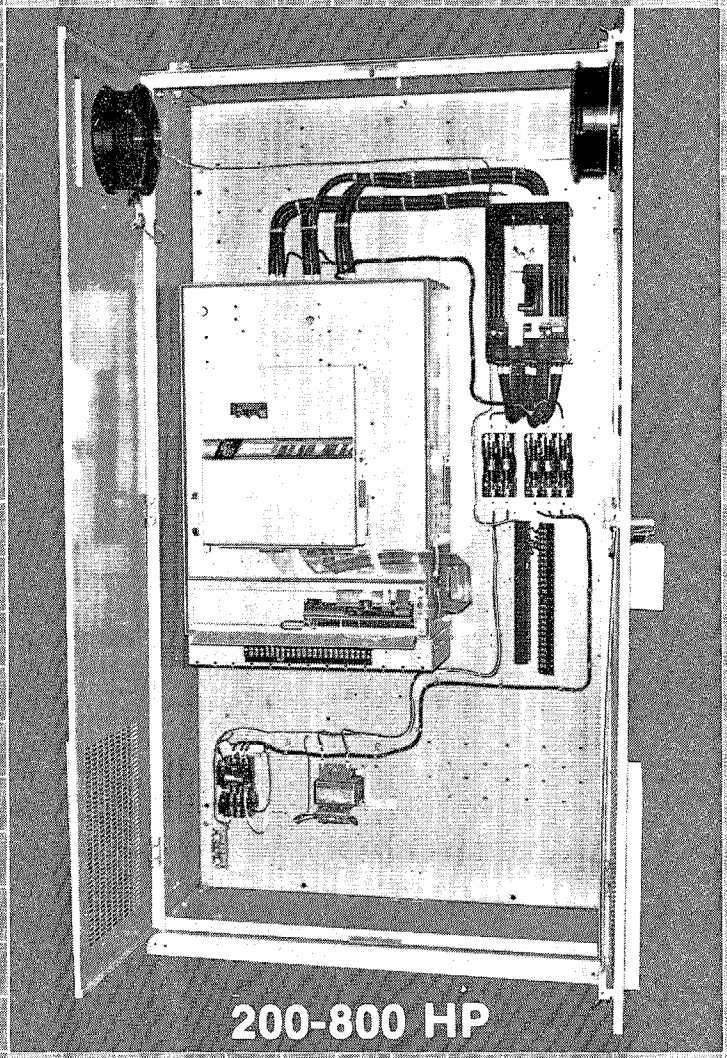
DC-200™ Adjustable Speed Drives



Installation, Operation and Maintenance Instructions

GENERAL ELECTRIC

TM TRADEMARK OF GENERAL ELECTRIC COMPANY, U.S.A.

INSTRUCTIONS

DC-200™ Adjustable Speed Drives

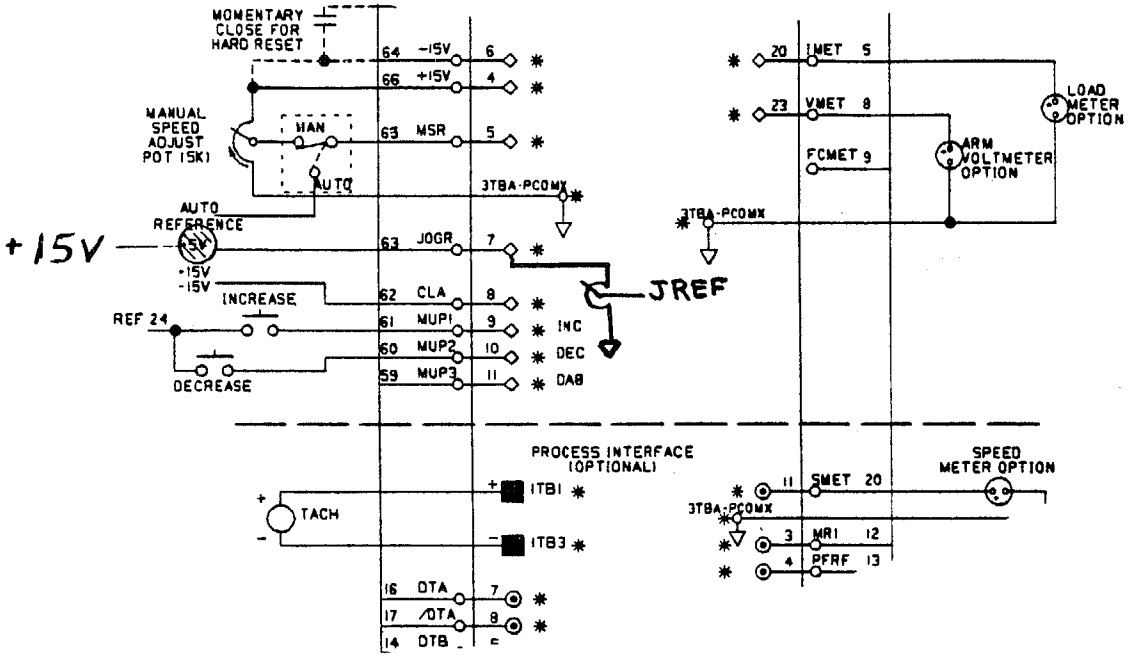
GEK-85770
ERRATA SHEET NO. 01
OCTOBER, 1987

This Errata Sheet affects GEK-85770. It should be attached inside the front cover and retained as a portion of that publication. The book's text should be changed in accordance with the following information:

Page 43 — Figure 20 Elementary Diagram — Sheet 2

The input to "JOGR" (3TB-63) must be from +15V, not +5V.

The +5V terminal (3TB-68) should not be connected to any external circuits.



These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

GENERAL ELECTRIC

GEK-85770
Errata Sheet No. 2
December, 1987

INSTRUCTIONS

DC-200 ADJUSTABLE SPEED DRIVES

This Errata Sheet affects GEK-85770, it should be attached inside the front cover and retained as a portion of that publication. The book's text should be changed with the following information.

ADD TO PAGE 23 — POWER CONNECTIONS

The DC-285 drive is equipped with a static motor field exciter that provides a fixed, unregulated DC output voltage for developing motor field excitation current. The actual field current is a result of the motor field voltage divided by the field resistance. In most cases this will allow satisfactory motor operation at rated base speed and below.

In some cases, due to low field resistance or high AC input voltage, the resulting motor field currents may exceed the maximum rated base speed field current. It, therefore, may not be possible to achieve rated base speed at full armature voltage. A measurement of the motor field current should be made. If it is found that the field current is above rated maximum it may be necessary to add a trim resistor in series with the field winding. The resistance can be approximated by measuring field current and field voltage at normal operating conditions and using the following calculations.

$$R(trim) = R(total) - R(field)$$

Where. . .

R(trim) is approximate resistance to be addeded in series with the motor field.

R(total) is the resulting total resistance required to achieve rated field current.

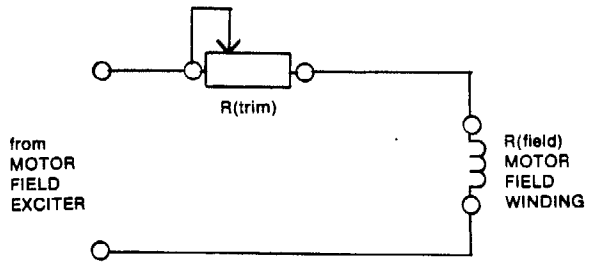
R(field) is the actual field resistance at operating temperature. (Note that this value may be different from the nominal cold resistance given on the motor rating plate)

Using Ohms Law and actual measured current and voltage, R(trim) may be approximated as. . .

$$R\ (trim) = \frac{\text{measured V (field)}}{\text{rated I (field)}} - \frac{\text{measured V (field)}}{\text{measured I (field)}}$$

Where the rated I (field) is the required maximum field current for base speed operation.

Using an adjustable resistor will allow the field current to be fine tuned. The Resistor should be sized per the maufacturers recommendations for the maximum field current and voltage.



These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

GENERAL ELECTRIC

TABLE OF CONTENTS

TITLE	PAGE
INTRODUCTION	3
RECEIVING HANDLING & STORAGE	3
SAFETY FOR PERSONNEL AND EQUIPMENT	4
INSTALLATION	4
MOUNTING	6
CONNECTIONS	6
MAINTENANCE	7
GENERAL DESCRIPTION	7
SPECIFICATIONS FOR BASIC DRIVES	10
MOTOR CONTROLLER ELEMENTS AND THEIR FUNCTION	11
AC LINE FUSES	11
FERRITE CORE ASSEMBLY	11
MAIN CONTACTOR (MD)	11
DC FUSE	11
CONTROL POWER TRANSFORMER (CPT)	11
CURRENT TRANSFORMER ASM (CT)	11
CONTROL CARD	11
POWER CONNECTION CARD	11
CONNECTIONS — POWER/CONTROL	23
START-UP PROCEDURE	26
HARDWARE JUMPER AND POT ADJUSTMENT	30
TEST POSTS	33
TROUBLESHOOTING	34
SPARE AND RENEWAL PARTS	47
APPENDIX I	52
APPENDIX II	56

LIST OF TABLES

FIGURE		PAGE
1	Relationship Between Air Temperature, Relative Humidity & Dew Point	3
2	Power Ratings	10
3	Speed vs Torque	10
4	Jumpers, Switches, Pots	31

ILLUSTRATIONS

FIGURE		PAGE
1	DC-200 Controller — Control/Power Connections	8
2	DC-200 Drive Controller Assembly	9
3	DC-200 Drive Controller With Driver/Regulator Door Open	12
4	DC-200 Block Diagram	13
5	DC-200 Snubber Card	14
6	MFE Power Circuit	15
7	Motor Field Exciter Assembly Location	16
8	Motor Field Exciter Assembly	17
9	Connection Diagram—Regenerative Controller	18
10	Connection Diagram—Non-regenerative Controller	18
11	Information Label Diagram—Non-regenerative Controller	19
12	Information Label—Regenerative Controller	20
13	Drive Control Card	21
14	Auxiliary Adjustment Card	21
15	Terminal Board Card	21
16	Power Connection Card—Non-regenerative Drive Controllers	22
17	Power Connection Card—Regenerative Drive Controllers	22
18	Typical Data Sheet	29
19	Elementary Diagram Sheet 1	42
20	Elementary Diagram Sheet 2	43
21	Voltage Feedback (VFB) At Discontinuous Current	44
22	Current Feedback (IMET) At Discontinuous Current	44
23	Voltage Feedback (VFB) At Continuous Current	44
24	Current Feedback (IMET) At Continuous Current	44
25	Gate Pulses With Gate Connected	44
26	Gate Pulses With Gate Disconnected	44
27	Dimensions—Controller only	45
28	DC-200 SCR Stack Assembly	50
29	DC-200 SCR Assembly	51
30	SCR Location—General	53
31	SCR Location/Fuse Data Non-regenerative	54
32	SCR Location/Fuse Data-Regenerative	55

INTRODUCTION

This Instruction Book contains helpful suggestions for placing the DC-200 drive equipment in service. It contains general information about drive operation and maintenance.

The operator and person responsible for maintaining this equipment should at **all times** have access to a copy of this instruction book and should consult this instruction book prior to operation or maintenance of the equipment.

RECEIVING HANDLING AND STORAGE

RECEIVING

The equipment should be placed under adequate cover immediately upon receipt as packing cases are not suitable for out-door or unprotected storage. Each shipment should be carefully examined upon arrival and checked with the packing list. Any shortage or damage should be reported promptly to the carrier. If required, assistance may be requested from **GENERAL ELECTRIC COMPANY, SPEED VARIATOR PRODUCTS OPERATION, ERIE, PA.** When seeking assistance please use drive serial number to identify the equipment.
Telephone 814-455-3219.

HANDLING

Wall mounted power units can be transported by lift trucks with the forks completely under the base, care being taken that the unit does not tip.

STORAGE

If the equipment is not to be installed immediately, it should be stored in a clean, dry location with ambient temperatures between —20°C (—4°F) and +55°C (131°F). The surrounding air must be free of chemical and electrically conductive or corrosive contaminants.

Precautions should be taken to prevent condensation from forming within the equipment enclosure. If the storage environment exceeds a 15°C. (27°F) drop in temperature at 50% humidity over a 4-hour period, a space heater should be installed inside each enclosure to prevent condensation (A 100 watt lamp can sometimes serve as a substitute source of heat.) Higher humidities with smaller temperature changes will also cause condensation.

Condensation occurs when air containing some moisture is cooled below its dew point. The dew point represents saturation of the air, and is the temperature at which the moisture starts to condense into water. It is not a fixed temperature but rather is related to the initial temperature of the air and its relative humidity at that temperature. The amount of moisture that can be held in the air is related to the air temperature. The following example illustrates some of these relationships.

In industrial drives, condensation is a possibility in applications where air temperature changes are large and rapid and/or the air is moist. For example, an outdoor crane operating in sunshine on a winter day, which then is shut down and parked in the shade will experience a rapid drop in temperature. This can result in condensation inside the equipment. Adding heat to keep the air temperature above its dew point can prevent condensation.

If storage temperatures below —20°C (—4°F) are likely to be present then auxiliary heat should be added in each enclosure to maintain temperature at or above —20°C. For assistance in heater size selection, contact General Electric Company.

When a drive that has been in operation is shut down for either a short or extended period of time, it is recommended that the environmental conditions be maintained the same as when in operation. Power unit ventilation or heating and air conditioning (if used) should be left on during the downtime to prevent large changes in temperature and possible moisture condensation.

TABLE I
Relationships Between Air Temperature, Relative Humidity and Dew Point

AIR TEMP °F	AIR TEMP °C	RELATIVE HUMIDITY %	WGT. OF MOISTURE IN 1 LB. OF DRY AIR GRAINS	DEW POINT °F	DEW POINT °C
104	40	100	345	104	40
104	40	80	270	97	36
104	40	40	130	75	24
104	40	10	32	37	3
50	10	100	54	50	10
50	10	80	42	43	6
50	10	40	21	25	4

SAFETY FOR PERSONNEL AND EQUIPMENT

The following paragraphs list some general safety reminders and safety recommendations to be followed when operating or installing this equipment.

WARNINGS, CAUTIONS, AND NOTES AS USED IN THIS PUBLICATION

WARNING

DENOTES OPERATING PROCEDURES AND PRACTICES THAT MAY RESULT IN PERSONAL INJURY OR LOSS OF LIFE IF NOT CORRECTLY FOLLOWED.

WHEN A WARNING LABEL IS PROVIDED IT WILL BE RED IN COLOR WITH BLACK OR WHITE LETTERING.

CAUTION

DENOTES OPERATING PROCEDURES AND PRACTICES THAT, IF NOT STRICTLY OBSERVED, MAY RESULT IN DAMAGE TO, OR DESTRUCTION OF, THE EQUIPMENT.

WHEN A CAUTION LABEL IS PROVIDED IT WILL BE AMBER IN COLOR WITH BLACK LETTERING.

NOTE

NOTES MERELY CALL ATTENTION TO INFORMATION THAT IS ESPECIALLY SIGNIFICANT IN UNDERSTANDING AND OPERATING THE EQUIPMENT.

WHEN A NOTE LABEL IS PROVIDED IT WILL BE WHITE IN COLOR WITH BLACK LETTERING.

WARNINGS

ALWAYS READ THE COMPLETE INSTRUCTIONS PRIOR TO APPLYING POWER OR TROUBLE-SHOOTING THE EQUIPMENT. FOLLOW THE START-UP PROCEDURES STEP BY STEP.

READ AND HEED ALL **WARNING, CAUTION AND NOTE** LABELS POSTED ON THE EQUIPMENT.

IMPROPER LIFTING PRACTICES CAN CAUSE SERIOUS OR FATAL INJURY.
LIFT ONLY WITH ADEQUATE EQUIPMENT AND TRAINED PERSONNEL.

WARNINGS

ELECTRIC SHOCK CAN CAUSE PERSONAL INJURY OR LOSS OF LIFE. CIRCUIT BREAKERS, IF SUPPLIED AS PART OF THE TOTAL SYSTEM, MAY NOT DISCONNECT ALL POWER TO THE EQUIPMENT. SEE SYSTEM ELEMENTARY DIAGRAMS. WHETHER THE AC VOLTAGE SUPPLY IS GROUNDED OR NOT, HIGH VOLTAGE TO GROUND WILL BE PRESENT AT MANY POINTS.

WHEN INSTRUMENTS SUCH AS OSCILLOSCOPES ARE USED TO WORK ON LIVE EQUIPMENT GREAT CAUTION MUST BE USED. THE OSCILLOSCOPE'S CHASSIS SHOULD BE GROUNDED AND A DIFFERENTIAL AMPLIFIER INPUT SHOULD BE USED. CARE SHOULD BE USED IN THE SELECTION OF PROBES AND LEADS AND IN THE ADJUSTMENT OF THE OSCILLOSCOPE SO THAT ACCURATE READINGS MAY BE MADE. SEE OSCILLOSCOPE MANUFACTURERS INSTRUCTION BOOK FOR PROPER OPERATION AND ADJUSTMENTS TO THE OSCILLOSCOPE.

SOME CONTROLLERS ARE FURNISHED WITH PARTIAL ENCLOSURES OPEN AT TOP AND BOTTOM. THESE ARE INTENDED ONLY FOR MOUNTING IN ANOTHER ENCLOSURE OR IN A CONTROL ROOM HAVING ACCESS BY QUALIFIED PERSONNEL ONLY.

EXPLOSIONS OR FIRES MIGHT RESULT FROM MOUNTING DRIVE CONTROLLERS IN HAZARDOUS AREAS SUCH AS LOCATIONS WHERE FLAMMABLE OR COMBUSTIBLE VAPORS OR DUSTS ARE PRESENT. DRIVE CONTROLLERS SHOULD BE INSTALLED AWAY FROM HAZARDOUS AREAS, EVEN IF USED WITH DC MOTORS SUITABLE FOR USE IN THESE LOCATIONS.

ALL MOTOR BASES AND EQUIPMENT ENCLOSURE HOUSINGS SHOULD BE CONNECTED TO THE FACTORY OR FACILITY EARTH GROUNDING SYSTEM.

CAUTIONS

DO NOT REMOVE CONNECTIONS OR PRINTED CIRCUIT CARDS FROM THE EQUIPMENT WHILE POWER IS APPLIED. THIS CAN DAMAGE THE EQUIPMENT.

DO NOT REMOVE INPUT POWER FROM THE DRIVE UNTIL IT HAS FULLY EXECUTED A STOP SEQUENCE AS THIS CAN DAMAGE THE DRIVE SYSTEM.

INSTALLATION WIRING MUST BE IN ACCORDANCE WITH THE NATIONAL ELECTRICAL CODE AND BE CONSISTENT WITH ALL LOCAL CODES. SECONDARIES OF 115 VOLT CONTROL TRANSFORMERS TYPICALLY HAVE ONE SIDE FUSED AND THE OTHER GROUNDED OR AVAILABLE FOR GROUNDING BY THE USER.

THE ELEMENTARY DIAGRAMS SUPPLIED WITH THE DC-200 DRIVE CONTROLLER IDENTIFY SOME SIGNAL CONNECTIONS AS "LOW LEVEL ANALOG AND/OR DIGITAL SIGNALS" AND INDICATE SPECIAL WIRING PRACTICES. THESE SIGNALS SHOULD BE ISOLATED FROM ALL OTHER WIRING (PARTICULARLY ANY POWER WIRES) AND RUN IN A SEPARATE CONDUIT OR WIRE RUN TO PREVENT POSSIBLE MISOPERATION OF THE CONTROLLER.

MEGGERING CAN DAMAGE ELECTRONIC COMPONENTS. DO NOT MEGGER OR HIPOT WITHOUT CONSULTING THE **SPEED VARIATOR PRODUCTS OPERATION, GENERAL ELECTRIC COMPANY.**

DO NOT CONNECT ANY EXTERNAL CIRCUITS OTHER THAN THOSE SHOWN ON THE ELEMENTARY DIAGRAM. CONNECTION OF DEVICES SUCH AS VOLTMETERS ON THE TACHOMETER MAY DEGRADE THE PERFORMANCE OF THE DRIVE SYSTEM.

DO NOT USE POWER FACTOR CORRECTION CAPACITORS WITH THIS EQUIPMENT WITHOUT CONSULTING THE **SPEED VARIATOR PRODUCTS OPERATION, GENERAL ELECTRIC COMPANY.** DAMAGE MAY RESULT FROM HIGH VOLTAGES GENERATED WHEN CAPACITORS ARE SWITCHED.

NOTES

IT IS RECOMMENDED THAT THE DRIVE SYSTEM COMMON (⏚) CIRCUIT BE GROUNDED (⏚) AT ONLY ONE POINT. THIS MEANS THAT IF THE DRIVE REFERENCE IS SUPPLIED BY A NUMERICAL CONTROL OR PROCESS INSTRUMENT WITH GROUNDED COMMON, THE DRIVE COMMON SHOULD NOT BE GROUNDED. IF A SEPARATE POWER TRANSFORMER IS USED AND IF THE SECONDARY OF THE TRANSFORMER MUST BE GROUNDED, IT IS RECOMMENDED THAT HIGH RESISTANCE GROUNDING BE USED.

INSTALLATION

LOCATION

DC-SCR drive power units are suitable for most factory areas where other industrial equipment is installed. They should be installed in well ventilated areas with ambient temperatures ranging from 0°C (32°F) to 40°C (104°F) and relative humidities up to 90 percent. It should be recognized, however, that since the life expectancy of any electronic component decreases with increased ambient temperature, reduction of the ambient temperature will bring about extended component life. For example, longer component life should be expected if the ambient temperature is held between 20°C (68°F) and 30°C (87°F).

Proper performance and normal operational life can be expected by maintaining a proper environment for the drive system.

Environments which include excessive amounts of one or more of the following characteristics should be considered hostile to drive performance and life:

1. Dirt, dust or foreign matter
2. Vibration or shock
3. Moisture or vapors
4. Temperature excursions
5. Caustic fumes
6. Power line fluctuations
7. Electromagnetic interference or "**NOISE**" introduced by —

   a. Radio frequency signals, typically portable transmitters used in the vicinity of the equipment or its wiring.

   b. Stray high voltage or high frequency signals as might be provided by arc welders, or by the operation of unsuppressed relay/contactor/brake coils as a part of or in the vicinity of drive control circuits.

WIRING LEVELS

1. Wiring should be run in separate conduits or wireways, for signal, control and power wiring levels defined as follows:

   a) Signal — Speed and/or position signals power supplies and logic signals (see system elementaries and notes to identify wire levels).

   b) Control — AC or DC control circuits

   c) Power — Armature and field leads Brakes, 115 VAC circuits

2) Signal wiring and power wiring may cross at right angles with a minimum one inch separation. Avoid parallel runs between signal and power wiring harnesses. If signal wire harnesses must be run in parallel to control or power wire harnesses, a minimum of a four-inch separation must be maintained.

GROUNDING

1) Drive common (PCOMX) should be grounded at **only one** point. If the reference is supplied by numerical control or process instrument with a grounded common the drive common should not be grounded separately.

2) If an isolation transformer is used and must be grounded, it is recommended that a high resistance ground be used.

3) Shields for shielded and twisted, shielded wire must be grounded on one end only. Provision is made to tie shields to chassis ground at the drive I/O. It is preferred to have the shield grounded only at the drive end.

COMMONS

Connection of reference, meter or any other external connection of commons can only be made at 3TB—PCOMX, unless otherwise indicated by the system elementaries.

SUPPRESSION

Control system relays, solenoids or brake coils can produce erratic drive behavior due to electrical noise transients. To eliminate this possibility, an RC suppressor should be added in parallel with the coils of these devices. A 220 ohm, 2 watt resistor in series with 0.5 mfd, 600 volt capacitor can be used.

TIGHTEN ALL CONNECTIONS

1) Incoming AC line connections.

2) Power connections to motor armature and motor field.

3) All terminal board (TB) connections including: 3TB, 4TB and 6TB.

MOUNTING

Totally enclosed controllers should be positioned to permit head radiation from all surfaces.

A wall mounted enclosure may be placed side by side with another enclosure. Clearance at least to the width of the enclosure doors should be available in front so that the door may be fully opened for easy access.

Wall mounted controllers may be mounted on any firm, reasonable flat, vertical surface.

CONNECTIONS

All internal electrical connections between components in the basic DC—SCR drive power units are made at the General Electric Company. Various modification options may be purchased, mounted and wired by the user or authorized General Electric distributor.

Be sure to protect the interior panel mounted components and sub-assemblies from metal particles when cutting or drilling entrances for interconnecting wiring and cables.

NOTE

OCCASIONALLY DRIVE ISOLATION TRANSFORMERS AND/OR OTHER ELECTRICAL COMPONENTS ARE SHIPPED SEPARATELY AND MUST BE MOUNTED AND CONNECTED TO THE SYSTEM.

NOTE

CONNECTION OF EXTERNAL CIRCUITS OTHER THAN SHOWN ON THE ELEMENTARY DIAGRAM, SUCH AS VOLTMETERS ON THE TACHOMETER, MAY DEGRADE THE PERFORMANCE OF THE DRIVE SYSTEM.

Before power is applied to the drive system, checks should be made to see that all internal connections are tight, and that all open relays and contactors operate freely by hand. Check that the equipment is clean and that no metal chips are present.

MAINTENANCE

Periodically inspect and maintain the equipment protective devices (particularly air filters when supplied) per instructions in this section. Check all electrical connections for tightness; look for signs of poor connections and overheating (arcing or discoloration).

DC MOTORS

Maintenance instructions covering brushes, commutator and lubrication are in GEH-2304 or GEH-3967 which are provided separately.

CAUTION

IT SHOULD BE NOTED THAT IF THE AC INPUT POWER IS NOT REMOVED FROM THE POWER UNIT THE MOTOR FIELD WILL CONTINUE TO BE EXCITED EVEN AT STAND STILL WITH THE MD CONTACTOR DROPPED OUT.

ALTHOUGH DC MOTORS ARE DESIGNED FOR FULL FIELD EXCITATION AT STAND STILL FOR PERIODS OF TIME (SEVERAL HOURS), UNDER THIS CONDITION THE MD CONTACTOR SHOULD BE DROPPED OUT, OTHERWISE THE FIELD COILS WILL BE SUBJECTED TO EXCESSIVE TEMPERATURE AND SIGNIFICANTLY REDUCED INSULATION LIFE.

WHENEVER THE MD CONTACTOR IS DROPPED OUT, A FIELD ECONOMY CIRCUIT WHICH AUTOMATICALLY REDUCES THE LEVEL OF EXCITATION IS EMPLOYED.

PRINTED CIRCUIT CARDS

Printed circuit cards normally do not require maintenance except to keep them clean and tightly secured. Clean as follows:

1. **Dry Dust** — Vacuum clean, then blow with dry filtered compressed air (low pressure supply).

2. **Oily Dirt** — Certain components (electrolytic capacitors, switches, meters, potentiometers and transformers) can be damaged by solvent, so use is not recommended. If absolutely necessary, clean using solvent sparingly and avoid above components. Clean contact terminals with dry non-linting cloth after solvent has been used. Recommended solvents: Freon* TE or TF.

3. If the card is badly contaminated or corroded, replace.

SILICON CONTROLLED RECTIFIERS

Keep SCRs and heatsink free from dirt, oil or grease, since any accumulation of dirt may cause overheating. Clean as follows:

1. **Dry Dust** — Vacuum clean, then blow with dry, filtered compressed air (low pressure).

CAUTION

SOLVENT CAN HARM NON-METAL COMPONENTS.

2. **Oily Dirt** — Use dry or barely moist (with solvent) non-linting cloth. Repeat until cloth remains clean. All SCRs must be cleaned with dry non-linting cloth after solvent has been used. Recommended solvents: Freon TE or TF.

CONTROL DEVICES

Inspect all relays and contactors at regular intervals and keep them free from dirt, oil or grease. Check for freedom of moving parts, corrosion, loose connections, worn or broken parts, charred insulation or odor, proper contact pressure and remaining wear allowance on contacts. Do not lubricate the contacts as lubrication shortens their life.

Both copper and silver contacts will become darkened and somewhat roughened in normal operation. This does not interfere with their performance, and does not indicate that the contacts should be filed. In general, contacts will not need attention during their normal life, but if prominent beads form on the surfaces due to severe arcing, the contact faces may be dressed with a fine file. **Do not** use sandpaper or emery cloth.

Any contact that is worn to the point where contact wipe or pressure is lost should be replaced. Cleaning procedure is the same as previously given for SCR heatsink.

GENERAL DESCRIPTION (Figure 2)

Power is fed through the AC fuses (or an optional disconnect switch), through current transformers and enters the power conversion SCR modules where it is converted to DC adjustable voltage. DC current is fed through a DC fuse (four quadrant controllers only) and MD contactor to the DC motor armature.

The basic elements of the DC-200 drive controller are shown in the simplified block diagram, Figure 4.

The speed of the motor is proportional to the DC voltage applied to its armature. Speed is measured and controlled by motor CEMF feedback (armature voltage feedback with **IR** compensation). As an optional feature, speed can be measured and regulated by a DC or AC tachometer, feedback.

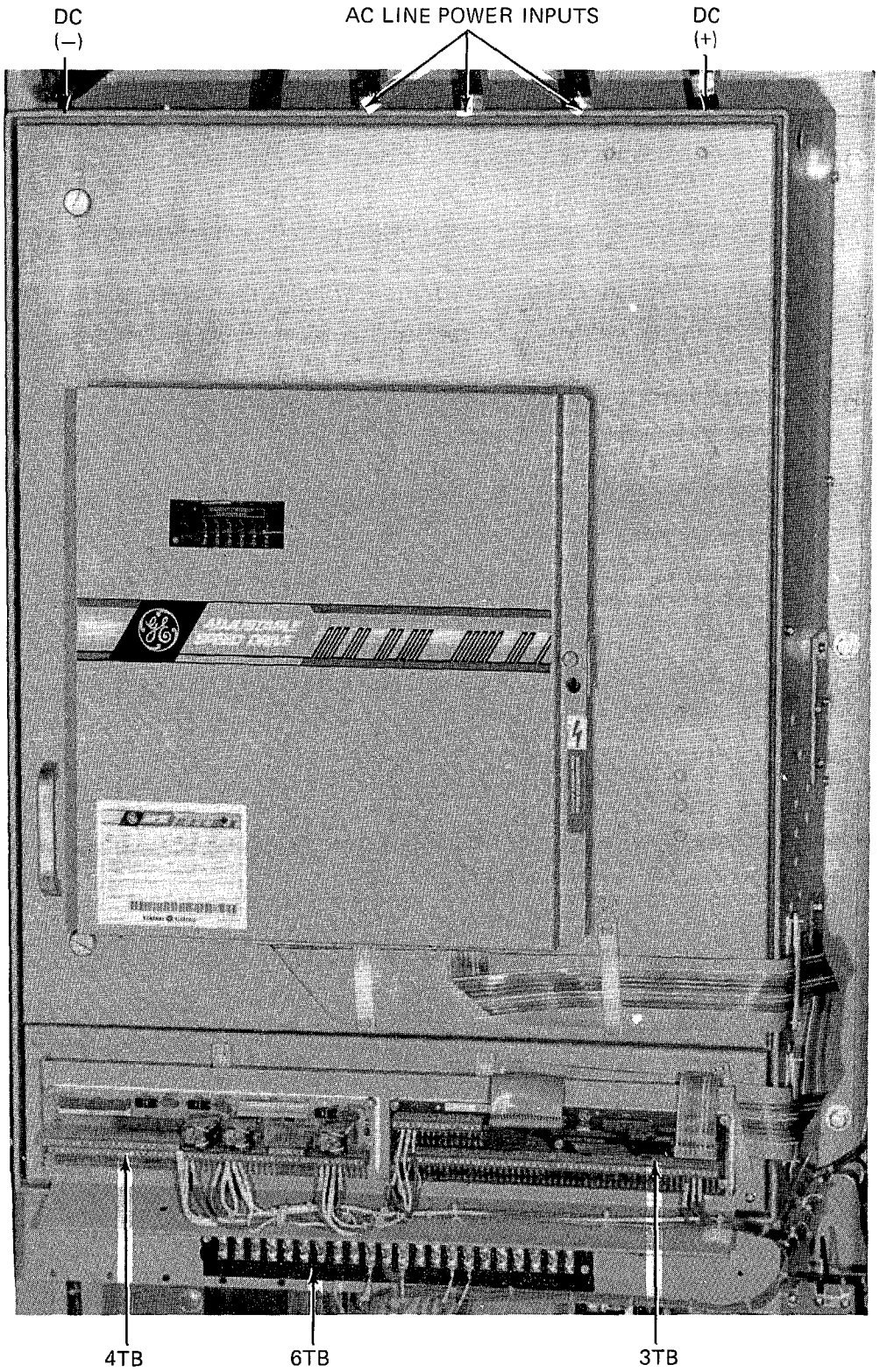


FIGURE 1
DC-200 CONTROLLER SHOWING
CONTROL AND POWER CONNECTIONS

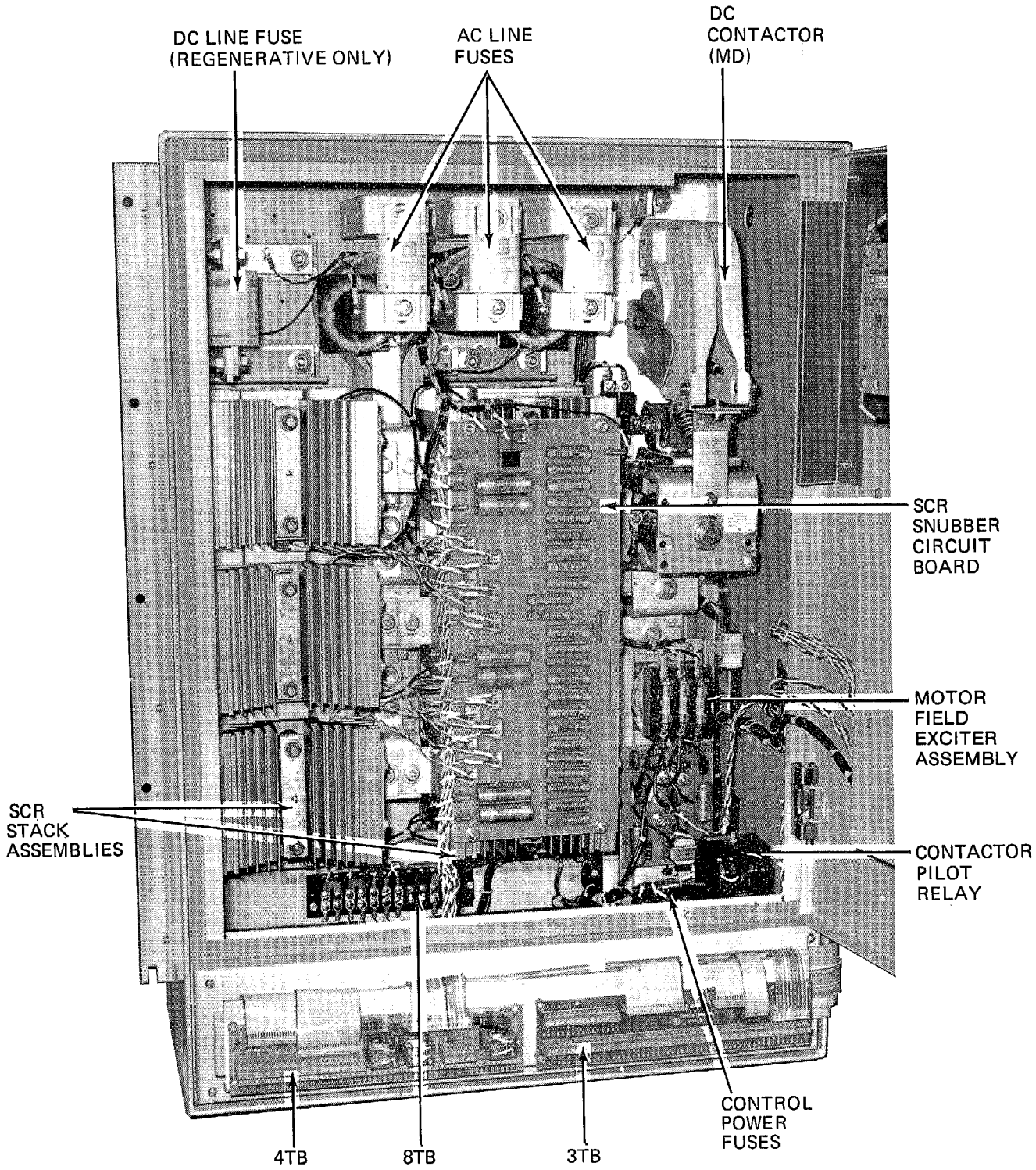


FIGURE 2
DC-200 POWER CONTROLLER ASSEMBLY

TABLE II
POWER RATINGS — 3 PHASE 50/60HZ

	AMPS - - INPUT/OUTPUT											
	230 VAC			460 VAC			575 VAC					
		OUTPUT			OUTPUT			OUTPUT			OUTPUT	
HP	INPUT LINE	240VDC ARM	150VDC FIELD	INPUT LINE	500VDC ARM	300VDC FIELD	INPUT LINE	630VDC ARM	300VDC FIELD	INPUT LINE	700VDC ARM	300VDC FIELD
125	359	440	24	—	—	—	—	—	—	—	—	—
150	428	525	24	—	—	—	—	—	—	—	—	—
200	543	666	24	—	—	—	—	—	—	—	—	—
250	698	855	24	—	—	—	—	—	—	—	—	—
300	837	1025	24	412	505	24	—	—	—	—	—	—
400	—	—	—	547	670	24	430	530	24	390	480	24
500	—	—	—	677	830	24	540	660	24	490	600	24
600	—	—	—	804	985	24	645	790	24	580	710	24
700	—	—	—	934	1145	24	735	900	24	661	810	24
800	—	—	—	1057	1285	24	850	1040	24	763	935	24
900	—	—	—	—	—	—	945	1160	24	847	1040	24
1000	—	—	—	—	—	—	1040	1280	24	938	1150	24

SERVICE CONDITIONS

- Rated Voltage — 230 VAC (-5, + 10%)
460 VAC (-5, + 10%)
575 VAC (-5, + 10%)
- Frequency — 60 Hertz (± 2 Hertz)
50 Hertz (± 2 Hertz)
- Ambient Temperature — 0° to 40°C (32° to 104°F)
- Altitude — Sea Level to 3300 feet
- Load Inertia — Not more than motor inertia when referred to motor shaft. For motor WK^2 refer to motor sheet.

CONTROLLED SPEED RANGE: 10:1

TABLE III SPEED VS TORQUE:

Typical data for speed versus torque:

MOTOR ENCLOSURE	Percent Rated Speed			
	100-60	50	40	5
	Percent Continuous Torque (Or Rated Amperes)			
DPFG (SELF VENT)	100	94	87	45
TENV or TEFC (SELF VENT)	100	95	92	77

MOTOR CONTROLLER ELEMENTS AND THEIR FUNCTION

[See Connection Diagrams (Figures 4, 5) As well as Information Labels (Figures 6, 7)]

AC LINE FUSES

AC line fuses (with or without an optional disconnect switch) are provided in the three incoming power lines to the controller and protect the SCRs, the wiring inside the controller and the wiring of the DC output.

FERRITE CORE ASSEMBLY (REGENERATIVE CONTROLLERS ONLY)

The ferrite core assembly is provided on the AC power lines between the AC line fuses and the SCRs to protect the SCRs from possible misoperation due to excessive rate of transient voltage change.

MAIN CONTACTOR (MD)

This contactor picks up when running mode is selected and no fault exists in the drive. This contactor drops out under the following conditions:

1. When a "Fault" condition occurs in the drive.
2. When the "Stop" command is selected.
3. When the drive regenerates to near zero speed after the "Stop" command (regenerative controllers only).

DC LINE FUSE* (REGENERATIVE CONTROLLERS ONLY)

A DC line fuse is provided in the DC output line to the motor armature and protects the SCRs and motor from overcurrent sourced from the motor.

CONTROL POWER TRANSFORMER (CPT)

This transformer is energized through its primary fuse located on the Control card. The secondary winding provides three isolated voltages:

1. 115 VAC to operate the coil of the main (MD) contactor and controller cooling fans.
   Do not use for external 115V excitation.
2. 38 VAC center-tapped for the Control card to provide DC control voltages necessary for proper controller operation.
   Do not use for external 38V excitation.
3. 16 VAC center-tapped for Control card to provide DC control voltages necessary for proper controller operation.
   Do not use for external 16V excitation.

CURRENT TRANSFORMER ASM (CT)

The current transformer assembly is used to provide armature current feedback signals to the control circuit.

MOTOR FIELD EXCITER ASSEMBLY

The motor field exciter assembly is used to provide excitation to the motor shunt field. The assembly contains the field power module (mounted on a heatsink), fuses and current transformer for the field circuit (see figures 6, 7 and 8). A field circuit suppression printed circuit board is also part of the assembly. Current feedback is connected to the control card by harness FCPL.

CONTROL CARD (See Figure 13)

This card provides the system intelligence for the drive controller. It contains one microprocessor and performs regulating, sequencing, SCR "burst" firing, diagnostic and protective functions for the drive along with many customer and motor interfacing functions. The functions performed on the card are implemented either in hardware and/or software. An additional function of the card is to display the fault status of the drive by means of 6 diagnostic LEDs mounted on the card. The LEDs which are illuminated are interpreted as a binary number indicative of the fault. See the troubleshooting section for a listing of the Fault codes.

This card provides the necessary ±15V, +5V and ±24V DC power used throughout the controller. This card also contains fuses for the power supply circuitry.

Three relays, "MAX" (K1), SLD1 (K3) and SLDM (K2) are on this card.

POWER CONNECTION CARD (See Figures 16 & 17)

This card contains SCR and DC output resistor-capacitor snubbers and the pulse transformers for the SCRs.

The DC voltage feedback isolation resistors are also mounted on this card.

SNUBBER CARD (See Figure 5)

The snubber card contains SCR resistor-capacitor snubber networks and the rectifier circuit for supplying DC coil power to the MD contactor.

***NOTE:** On Regenerative Controllers with paralleled DC fuses, replace both fuses, even if only one is blown.

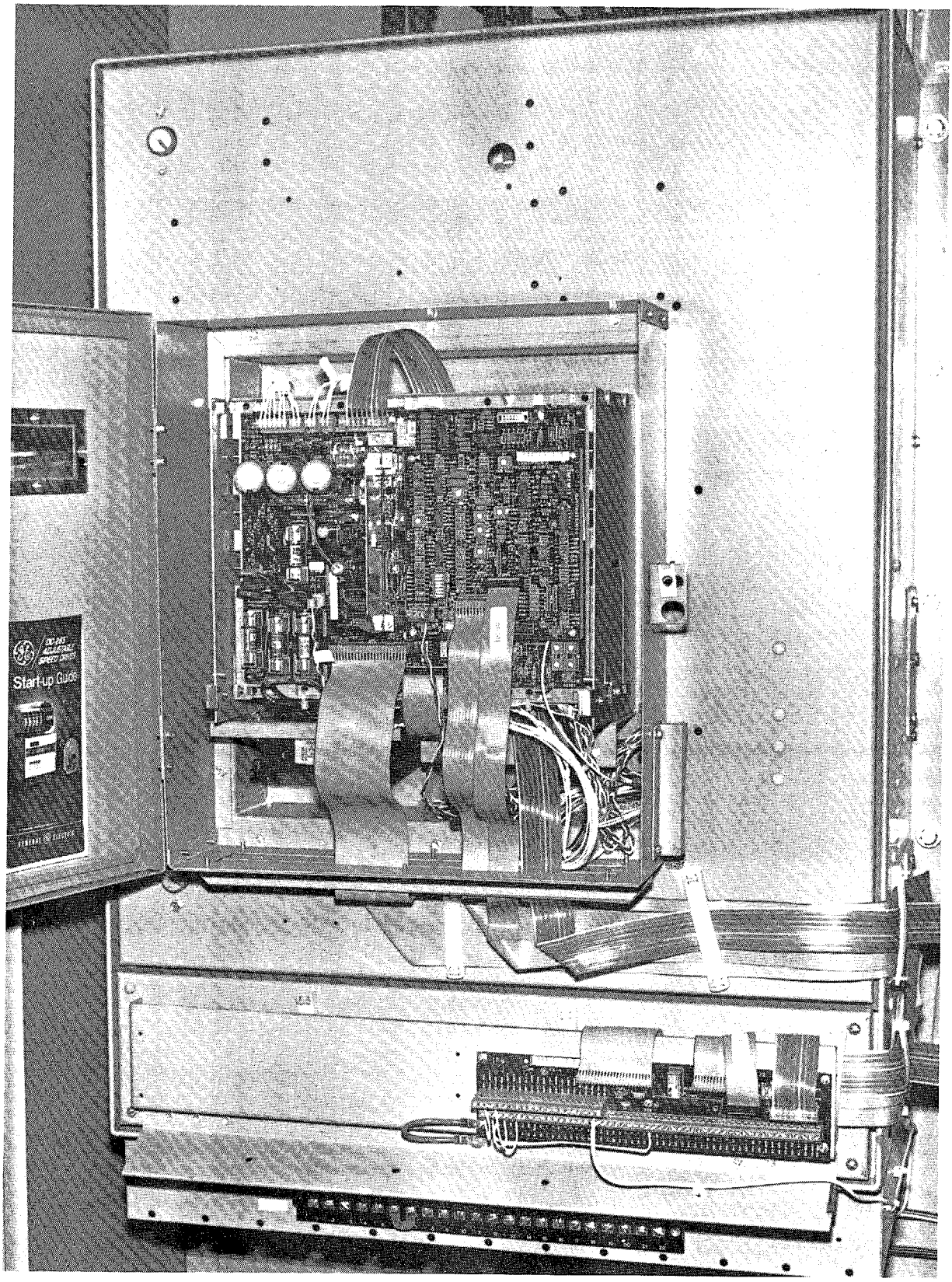


FIGURE 3
DC-200 DRIVE CONTROLLER
WITH DRIVER/REGULATOR

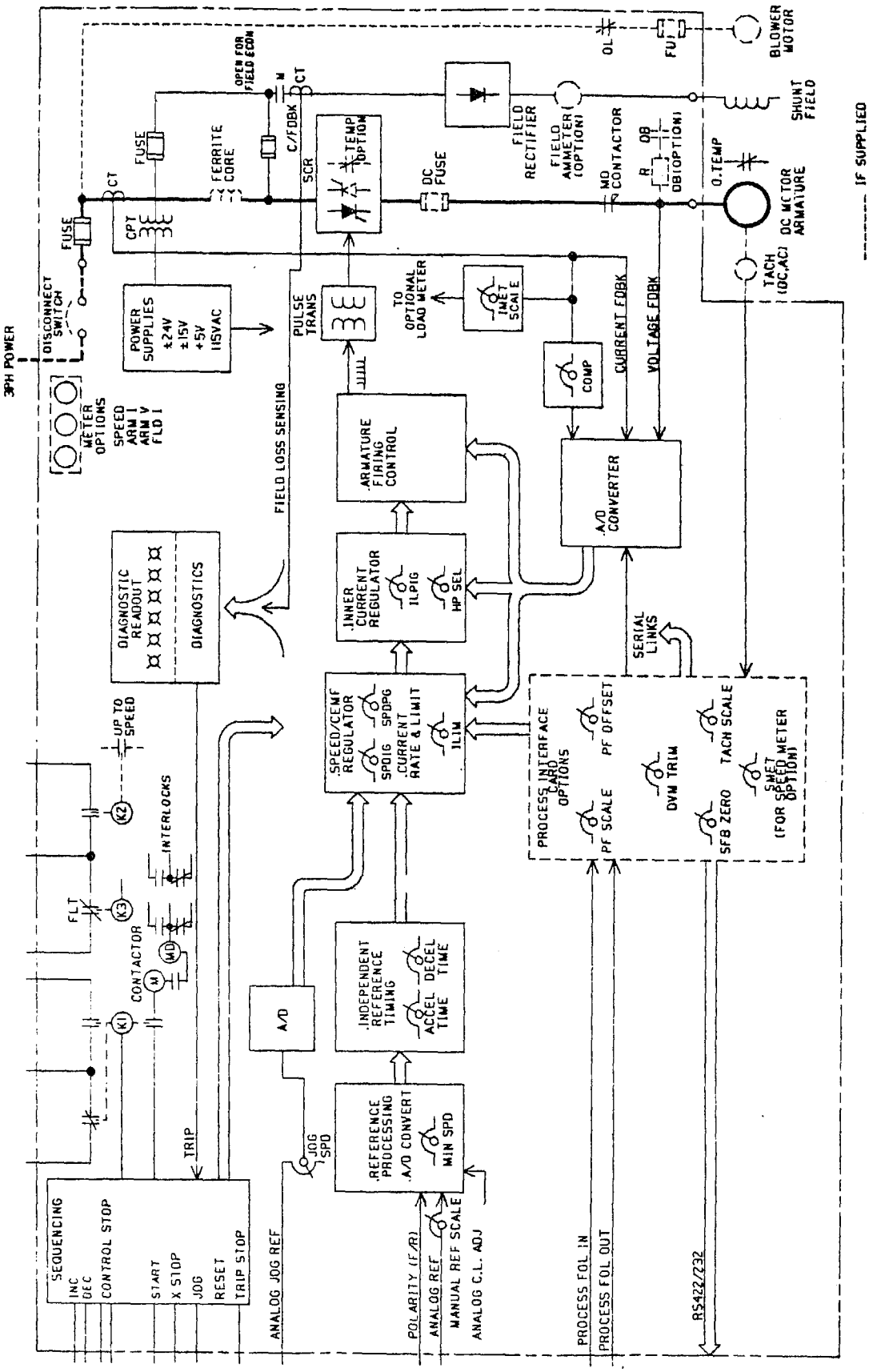


FIGURE 4
BLOCK DIAGRAM — DC-200 DRIVE CONTROLLER

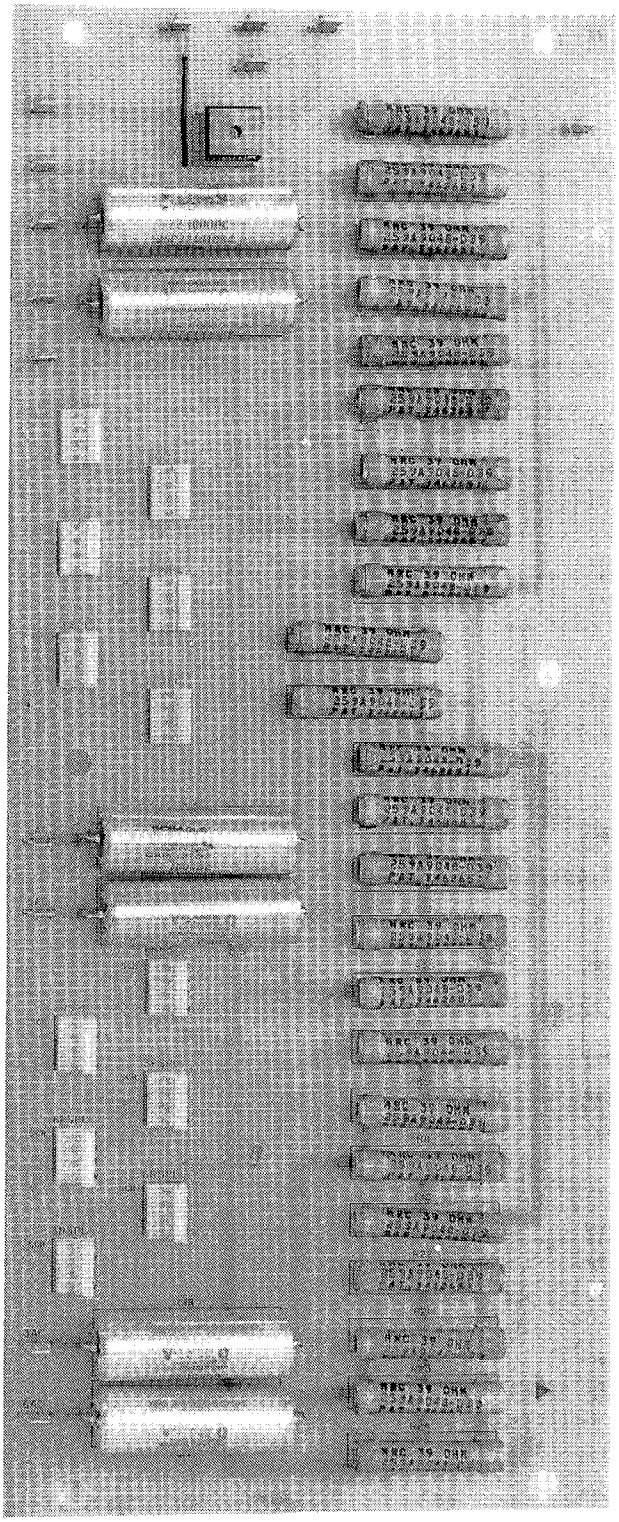


FIGURE 5
DC-200 SNUBBER CARD

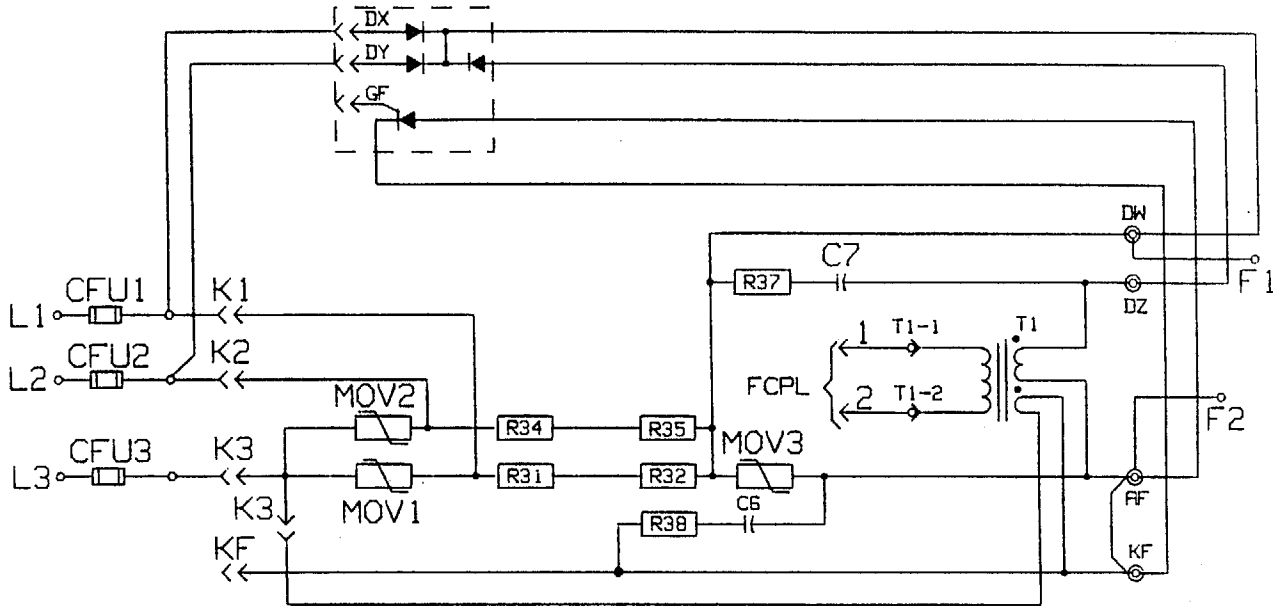


FIGURE 6
MFE POWER CIRCUIT

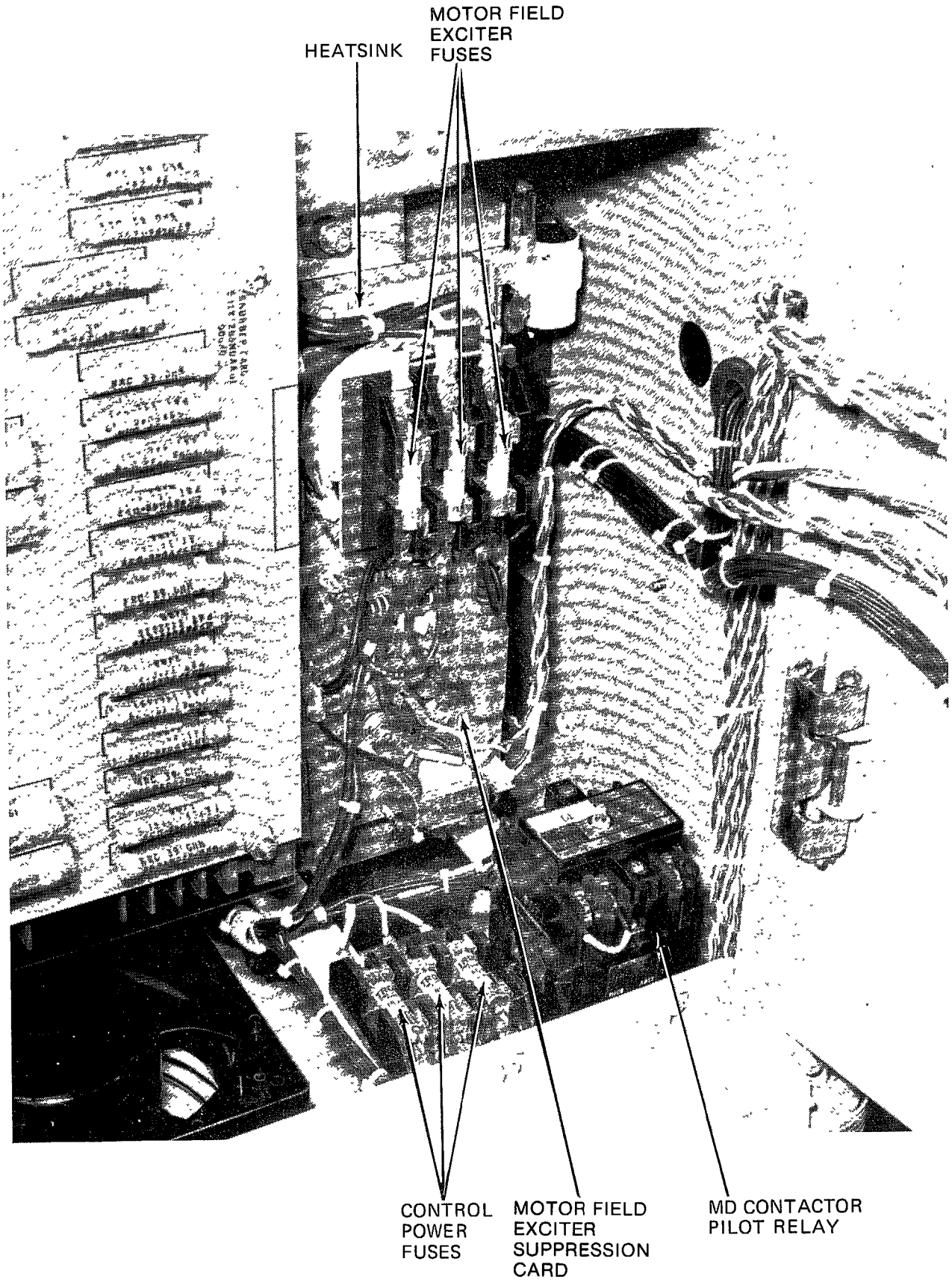


FIGURE 7
MOTOR FIELD EXCITER ASSEMBLY

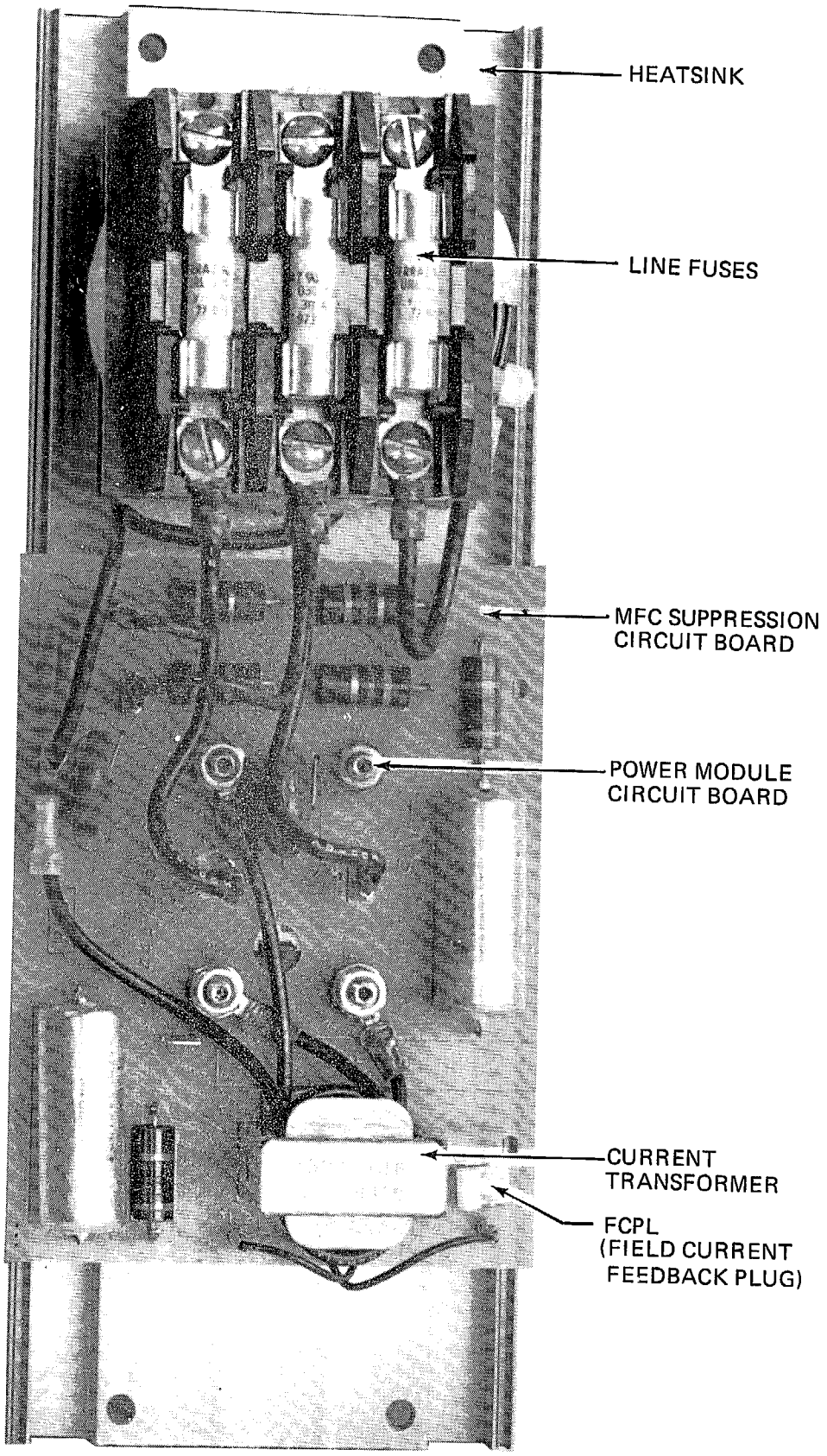


FIGURE 8
MOTOR FIELD EXCITER ASSEMBLY

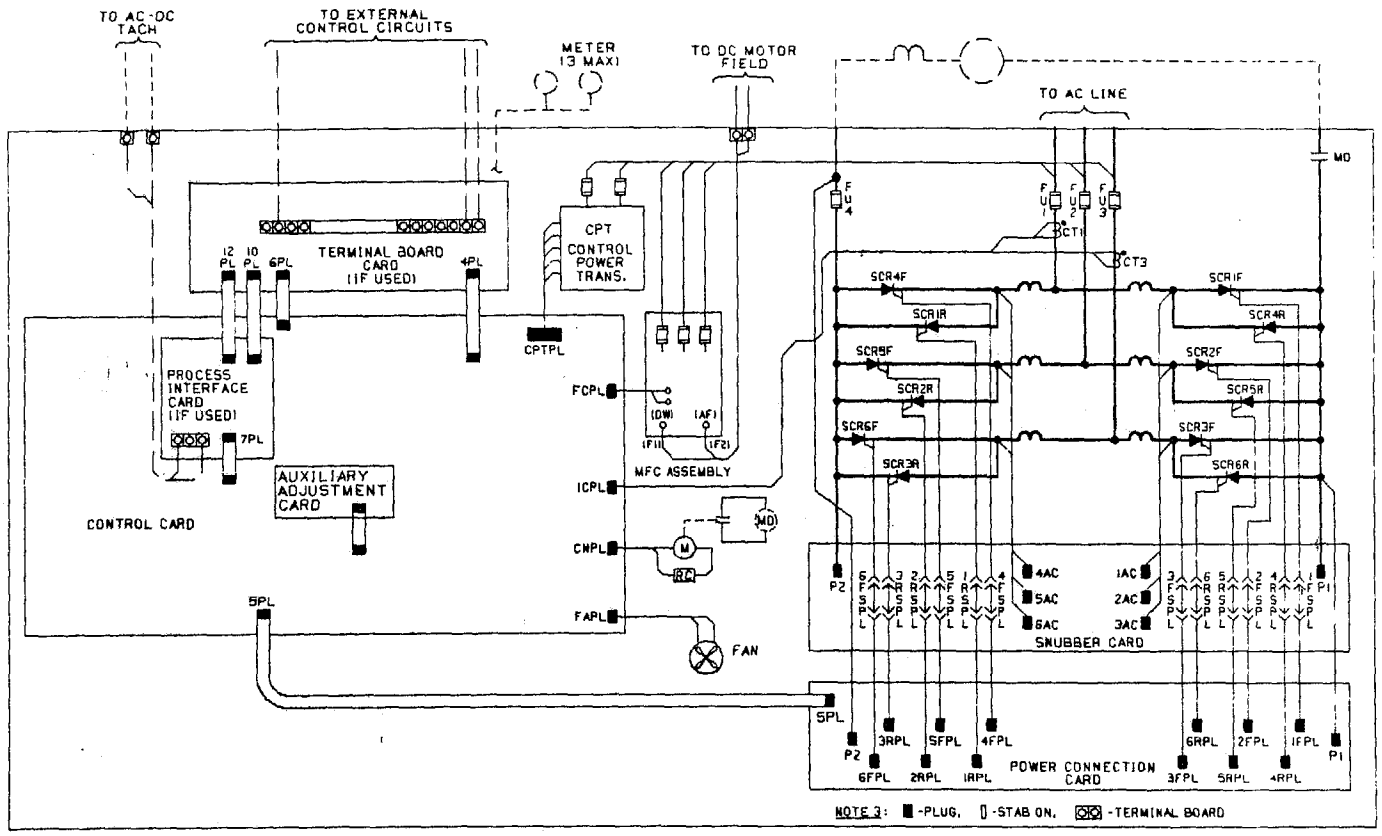


FIGURE 9 — CONNECTION DIAGRAM — REGENERATIVE CONTROLLER

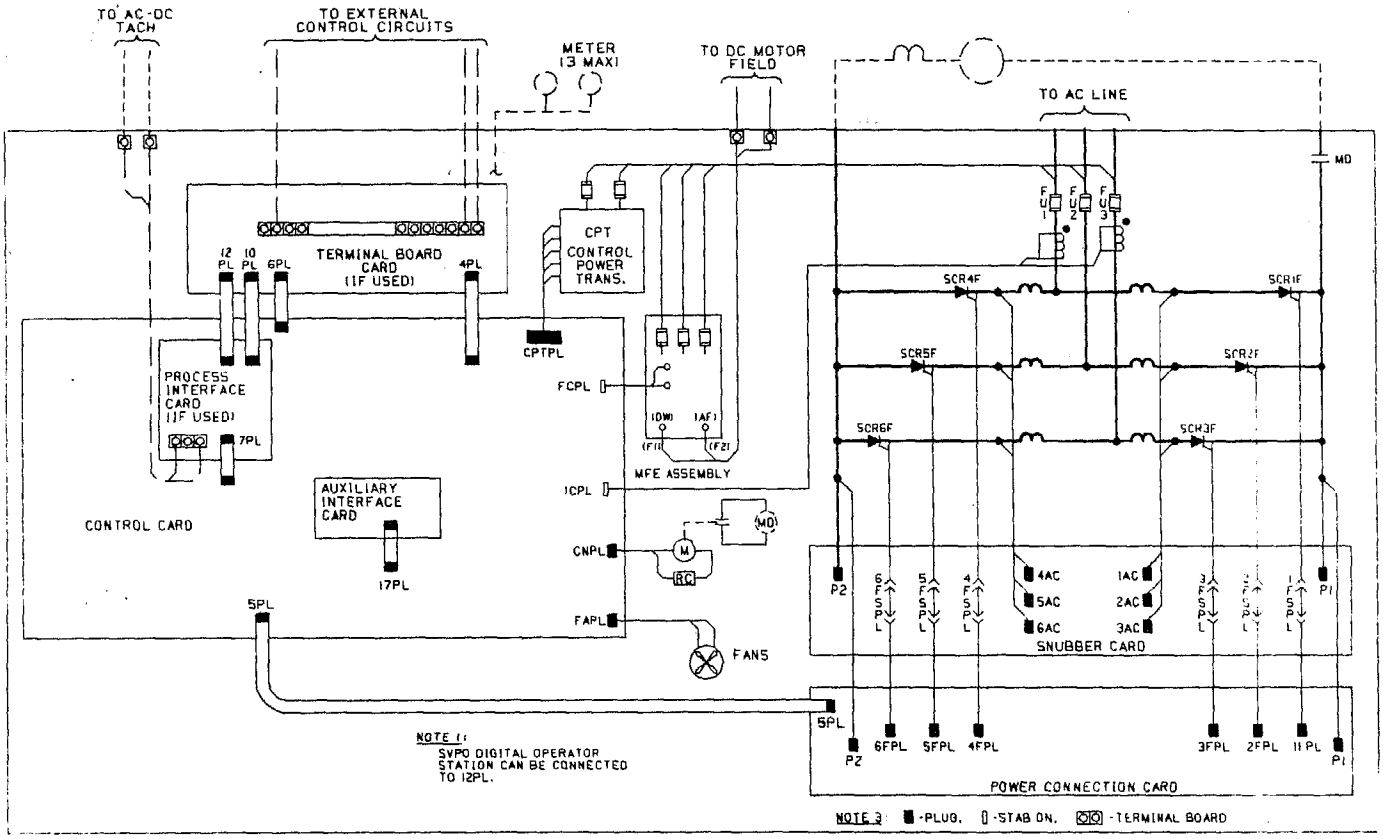
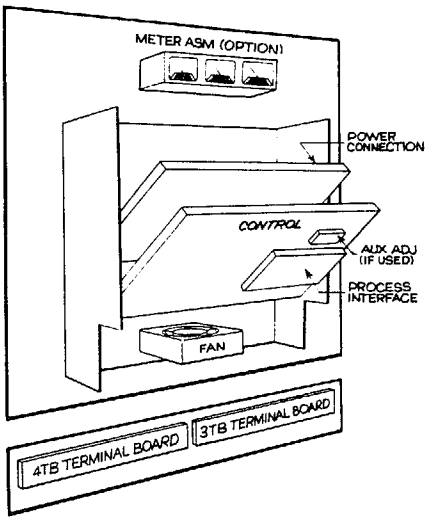


FIGURE 10 — CONNECTION DIAGRAM—NON REGENERATIVE CONTROLLER



For Service

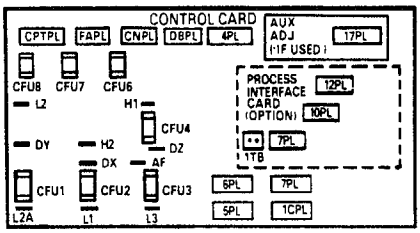
For Factory Service and Technical Information on this product, call Erie, Pennsylvania 814-455-3219 (24 Hour Phone Service) — or contact the nearest General Electric Apparatus & Engineering Services Office listed in your telephone directory. Before calling, list Model and Serial numbers of the power unit and motor.

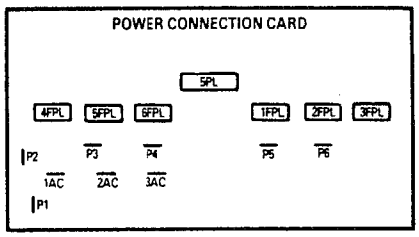
For Parts

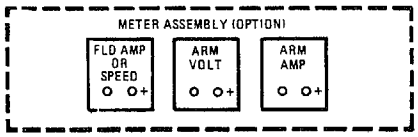
Component parts for this equipment can be obtained from your nearest General Electric Apparatus Sales representative or authorized General Electric Company distributor.

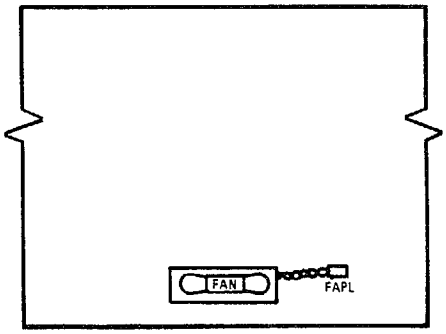
If needed parts cannot be identified by number or if they are not readily available through standard ordering procedures, request your representative to contact the Speed Variator Products Operation, Erie, Pennsylvania, giving Model number, ML number and Serial number of the complete equipment.

Additional information can be obtained from the equipment instruction book.





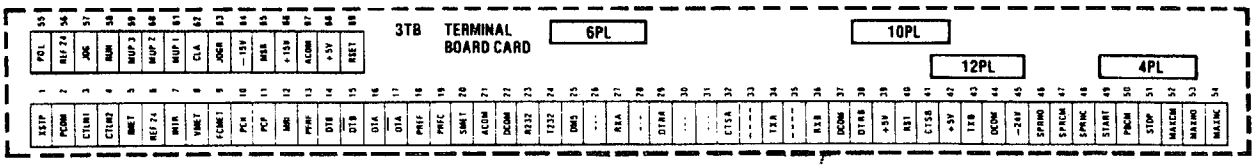


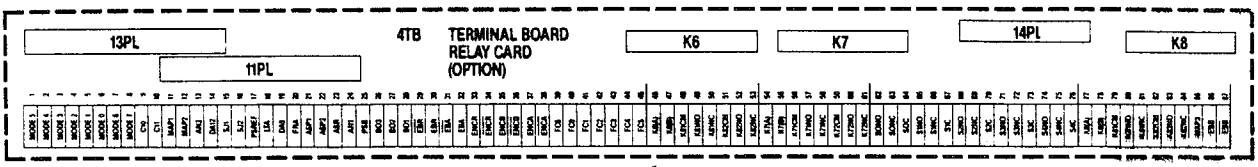


MOTOR FIELD CONTROL / POWER SUPPLY FUSE

Vac	NAME	A	V	TYPE
230 / 460	CFU1-3	15	600	ATM*
230 / 460	CFU4	3.2	500	FNQ**
230 / 460	CFU6	2.0	250	312***
230 / 460	CFU7	0.5	250	312***
230 / 460	CFU8	1.0	250	312***

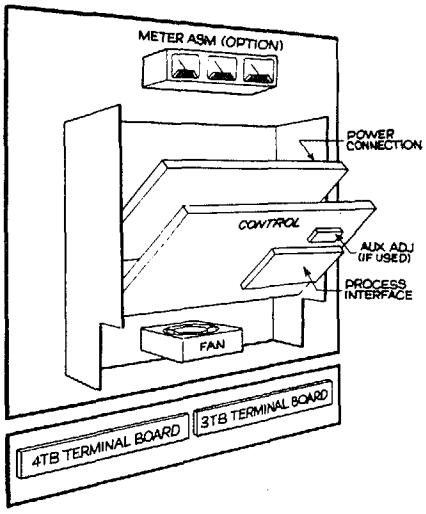
*Gould-Shawmut or Equivalent
**Bussmann or Equivalent
***Little Fuse or Equivalent





NP 104X9058A466 2/86

FIGURE 11
DC-200 INFORMATION LABEL
NON-REGENERATIVE DRIVE CONTROLLER



For Service

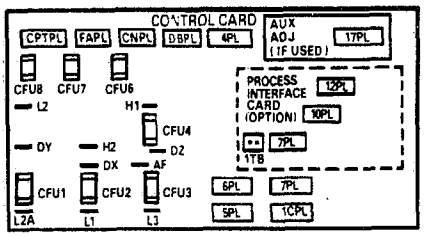
For Factory Service and Technical Information on this product, call Erie, Pennsylvania 814-455-3219 (24 Hour Phone Service) — or contact the nearest General Electric Apparatus & Engineering Services Office listed in your telephone directory. Before calling, list Model and Serial numbers of the power unit and motor.

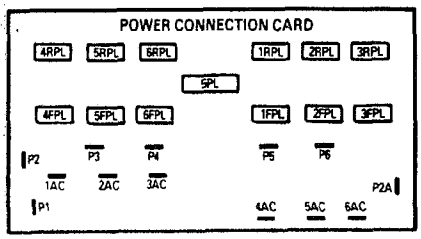
For Parts

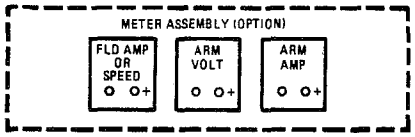
Component parts for this equipment can be obtained from your nearest General Electric Apparatus Sales representative or authorized General Electric Company distributor.

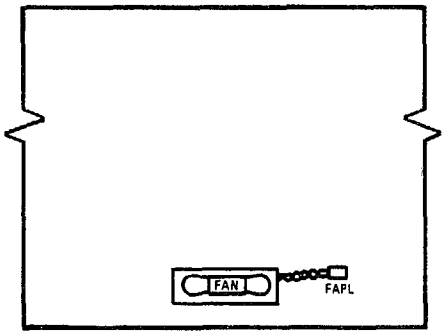
If needed parts cannot be identified by number or if they are not readily available through standard ordering procedures, request your representative to contact the Speed Variator Products Operation, Erie, Pennsylvania, giving Model number, ML number and Serial number of the complete equipment.

Additional information can be obtained from the equipment instruction book.



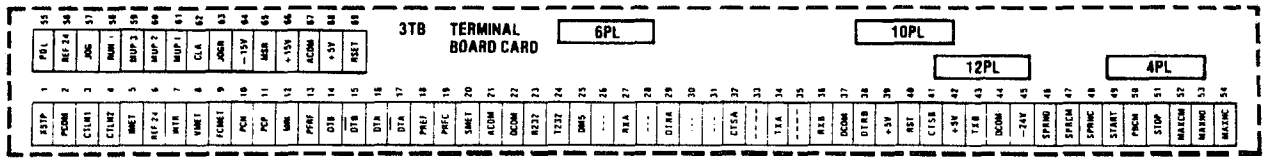


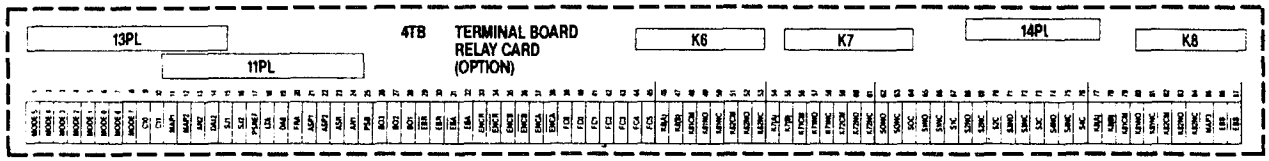




MOTOR FIELD CONTROL / POWER SUPPLY FUSE				
Vac	NAME	A	V	TYPE
230 / 460	CFU1-3	15	600	ATM*
230 / 460	CFU4	3.2	500	FNQ**
230 / 460	CFU6	2.0	250	312***
230 / 460	CFU7	0.5	250	312***
230 / 460	CFU8	1.0	250	312***

*Gould-Shawmut or Equivalent
**Bussmann or Equivalent
***Little Fuse or Equivalent





NP 104X9058A467 2/86

FIGURE 12
DC-200 INFORMATION LABEL
REGENERATIVE DRIVE CONTROLLER

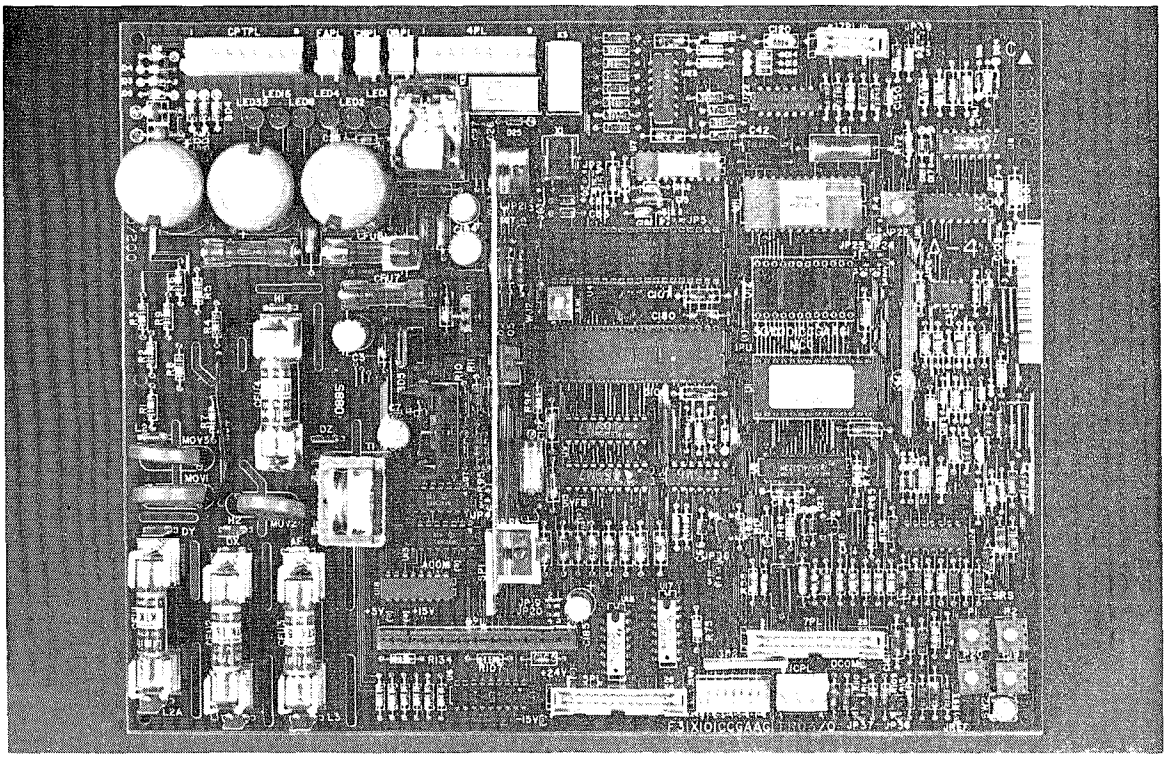


FIGURE 13 DC-200 DRIVE CONTROL CARD

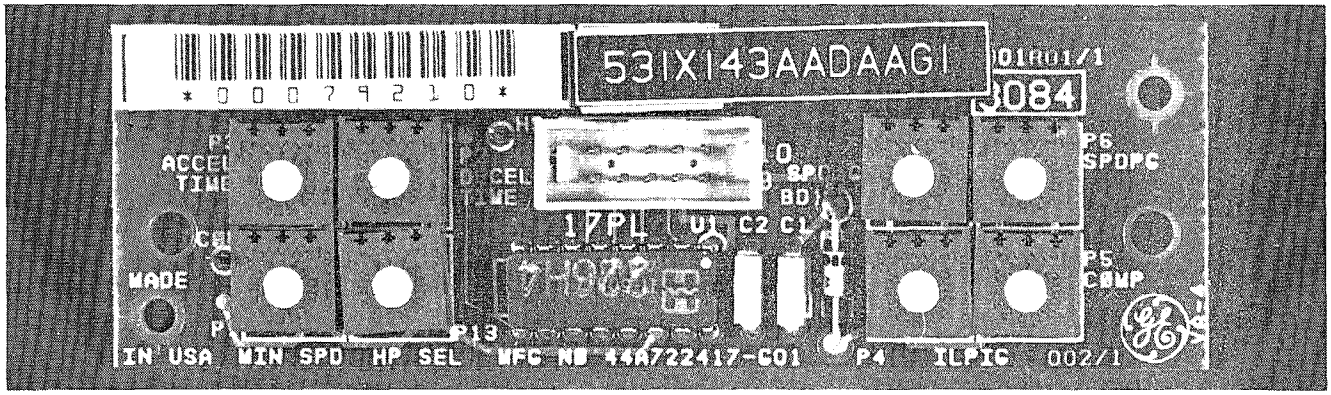


FIGURE 14 AUXULIARY ADJUSTMENT CARD

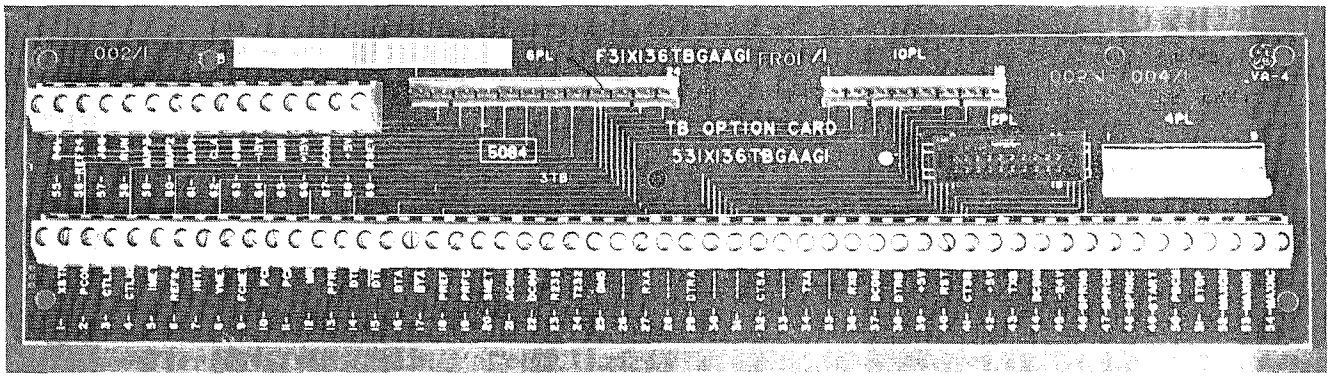


FIGURE 15 DC-200 TERMINAL BOARD CARD

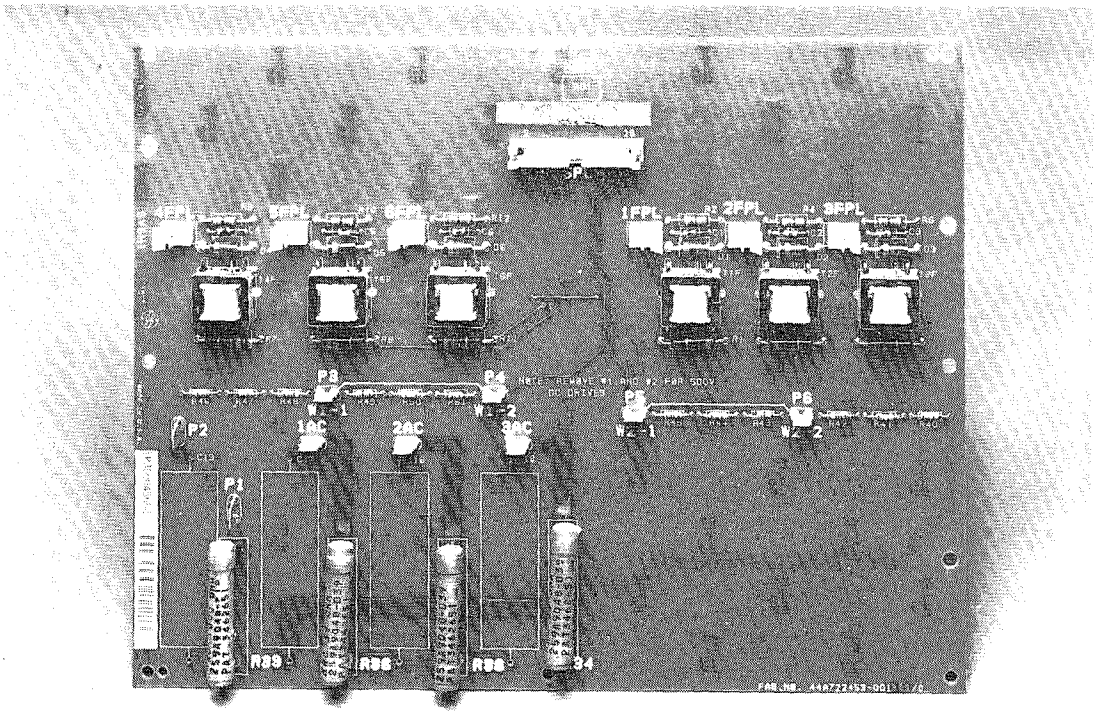


FIGURE 16
DC-200 POWER CONNECTION CARD
NON-REGENERATIVE DRIVE CONTROLLERS

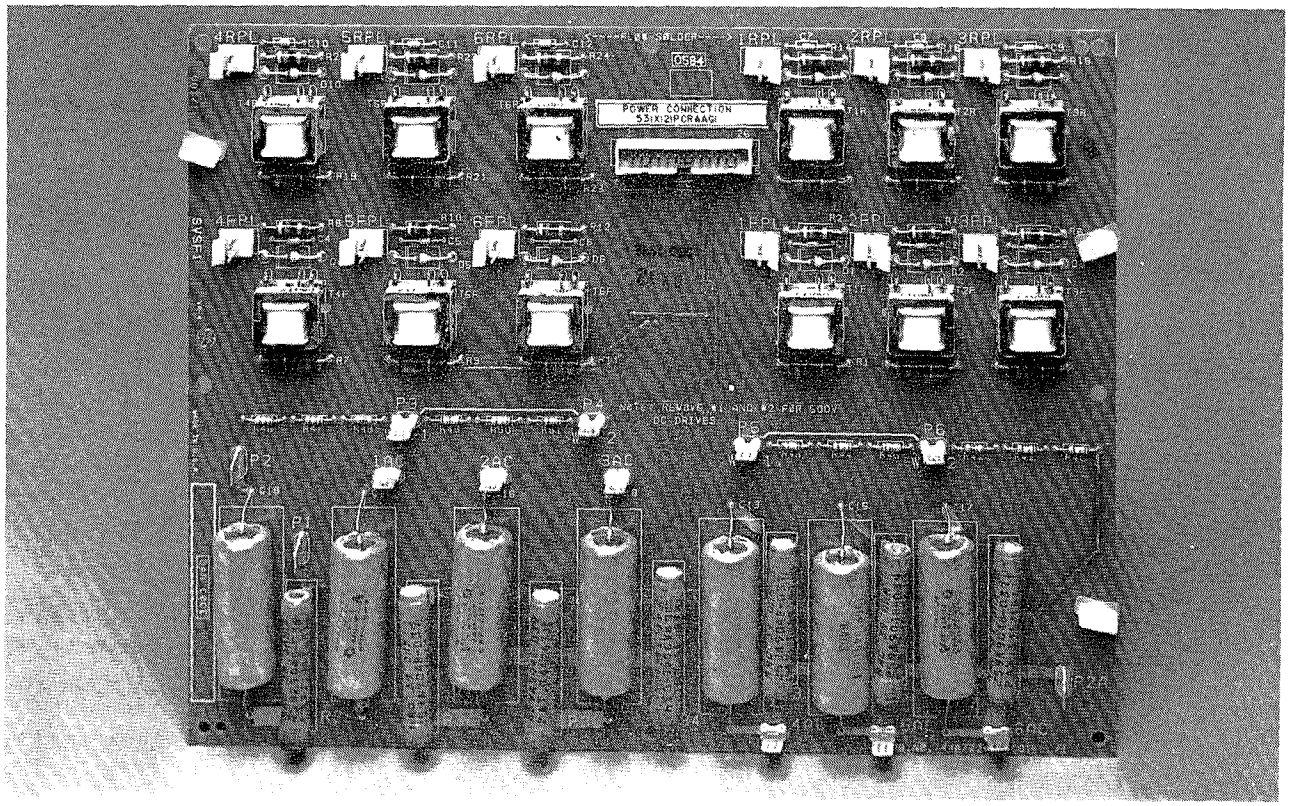


FIGURE 17
DC-200 POWER CONNECTION CARD
REGENERATIVE DRIVE CONTROLLERS

CONNECTIONS

POWER CONNECTIONS

The three phase input AC power cables are connected at L1, L2 and L3 on the line side of the AC fuses.

The DC motor shunt field leads are connected at 6TB4 (+) and 5 (-).

The DC motor armature cables are connected at P1 (+)at the top of the contactor and P2 (-) at the top of the DC fuse (FU4) and/or buss as shown in Figure 2.

CONTROL CONNECTIONS (See Figures 9 & 10)

The control wiring to the basic controller is terminated at 3TB on the Terminal Board card for all Control card and most Process Interface card connections. Points are referenced by both nomenclature and number. Additional connections may be made through 6TB:

6TB POINT CONTROL FUNCTION DESCRIPTION

1	230/460V, 3 phase input, used when non-standard input voltages are supplied at L1, L2, L3
2	
3	
4	Shunt Field Excitation (+)
5	Shunt Field Excitation (-)
6	External Fault Interlock
7	External Fault Interlock
8	Optional 115 VAC (.25kVA) output
9	Optional 115 VAC (.25kVA) output
10	SCR stack thermal switch output (Stack No. 1)
11	SCR stack thermal switch output (Stack No. 1)
12	SCR stack thermal switch output (Stack No. 2)
13	Tachometer Feedback input (+)
14	Tachometer feedback input (—)
15	MD Contactor Auxiliary Interlock: (15 to 16 = N.O.) (16 to 17 = N.C.) (18 to 19 = N.O.) (19 to 20 = N.C.)
16	
17	
18	
19	
20	
21	Optional MOV panel fuse loss interlock
22	Optional MOV panel fuse loss interlock

START/STOP/ CIRCUITRY OPERATION

The only allowable locations for Control card jumpers 19 & 20 (JP19 & JP20) are in the 1-2 and 3-4 positions respectively.

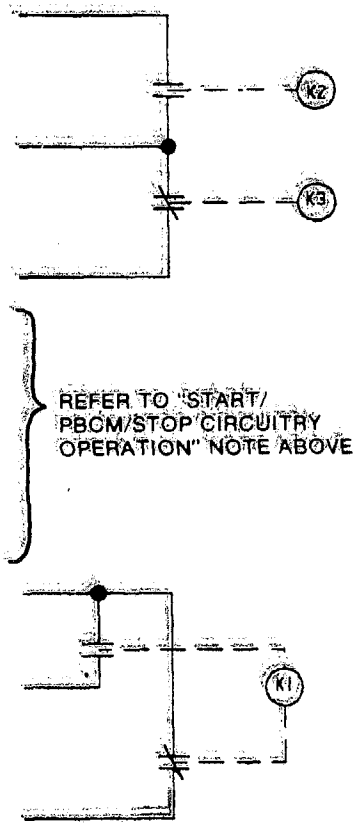
With the jumpers in this position, and with STOP connected to PBCM (or -24V), the START input is active (allows controller to run) when connected to PBCM (or-24V). When START is released, the controller will continue to run until STOP is released from PBCM (or -24V). STOP may then be returned to PBCM (or -24V) in preparation for the next start command.

If START is permanently connected to PBCM (or -24V) then a controller run will be initiated when STOP is connected to PBCM and will stop when STOP is released to PBCM.

The function of each connector and its pins is described in the following:

CONNECTOR 4PL — LOCATED ON CONTROL CARD

4PL PIN NO.	3TB NOMENCLATURE	DESCRIPTION
1	SPRNO (46)	Up to Speed Normally Open Contact
2	SPRCM (47)	Relay Common Connection
3	SPRNC (48)	Fault Relay Normally Closed Contact
4	START (49)	Connection Point for Start Pushbuttons
5	PBCM (50)	Common Point for Start and Stop Pushbuttons
6	STOP (51)	Connection for Stop Pushbutton
7	MAXCM (52)	MA Auxiliary Relay Common Connection
8	MAXNO (53)	MA Auxiliary Relay Normally Open Contact
9	MAXNC (54)	MA Auxiliary Relay Normally Closed Contact



Note: Refer to elementary diagrams for contact ratings.

CONNECTOR 6PL - LOCATED ON CONTROL CARD

PIN NO.	3TB NOMENCLATURE	DESCRIPTIONS
1	RSET (69)	A hardware (hard) reset of drive is initiated when RESET is connected to +15V and must be open to run. RSET immediately stops firing of all SCRs, drops delayed RESET SCR firing power, MD contactor, fault relay, and holds the microprocessor in a reset state until RESET is opened. The drive will not automatically restart if reset while running. Do not apply RESET while running. Do not apply RESET while regenerating.
2	+5V (68)	Not used.
3	ACOM (67)	Not used.
4	+15V (66)	Signal voltage for System Reference pot and RESET circuit.
5	MSR (65)	Manual system reference input to drive ± 15V max, positive for forward motoring with POL at zero or open.
6	-15V (64)	Signal Voltage for system reference pot.
7	JOGR (63)	Jog reference input (bypasses linear time), ±14.4V = Top speed if JOG SPD potentiometer is in the full CW position.
8	CLA(62)	Current limit adjust-adds to setting of current limit. +10V increases current limit by about 125% of set, -10V decreases it by about 125%, depending on ILIM pot setting.
9	MUP1 (INC) (61)	Control input that commands INCrement speed in the MRH mode.*
10	MUP2 (DEC) (60)	Control input that commands DECrement speed in the MRH mode*
11	MUP3 (DA8) (59)	Eight bit digital to analog current feedback signal. 2 volts = 100% current.
12	RUN (58)	Must be activated (pulled to REF24) to enable controller run. When released, a coast and trip stop is initiated.
13	JOG (57)	Jog command input point. Activated by connecting JOG to REF24. Controller stops when JOG is released.
14	REF24 (56)	±24V supply for XSTP, JOG, POL, REV, INC, DEC or INTR input.
15	POL (55)	Control input which reverses the direction of the drive for any applied reference; e.g, activating POL is equivalent to reversing the sign of the input at MSR. Activated connecting POL to REF 24.
16	XSTP (1)	Must be pulled to REF24 to enable controller. When released, a coast and trip stop is initiated.
17	PCOM (2)	Return line for meter drivers IMET and VMET.
18	CTLN1 (3)	Must be connected to CTLN2 to allow the drive to run. Opening this connection unconditionally drops MA contactor causing an uncontrolled stop. If the MA contactor is opened under load, harmful contact arcing may occur.

* To activate the MRH mode, the "INC" input must be pulled to REF24 while START is pulled to PBCM. Both inputs can then be released. The controller will continue to run and obey the "INC" and "DEC" commands until STOP is released from PBCM.

CONNECTOR 6PL — LOCATED ON CONTROL CARD (continued)

PIN NO.	3TB NOMENCLATURE	DESCRIPTION
19	CTLN2 (4)	See CTLN1.
20	IMET (5)	Analog load meter output proportional to armature current. See Appendix II for details on meter kits.
21	REF24 (6)	±24V supply for RUN, JOG, POL, XSTP, INC, DEC or INTR.
22	INTR (7)	SLD5 control signal.
23	VMET (8)	Analog armature voltmeter output. This output is 4 volts at rated armature volts output. Maximum load is 2 milliamps.
24	FCMET (9)	Not used.

START—UP PROCEDURE

(REVIEW WARNINGS AND CAUTION NOTICES BEFORE PROCEEDING)

Every DC-200 DC SCR drive controller has been factory tested and is ready to operate provided the power and control connections have been properly made and the following steps are followed.

(If the Digital Operator Station is connected through a Process Interface card, refer to instruction book in the Appendix II.)

The DC-200 is equipped with a 2/3 wave motor field exciter circuit. No internal adjustments can be made to vary the field strength. Field loss detection circuitry is included and is enabled by placing jumper JP40 in the 1-2 position. Field loss detection can be disabled by placing jumper JP40 in the 2-3 position.

1. Verify that all terminal board screws are tight.

2. Verify that the incoming power is the proper voltage and frequency and the incoming wiring is complete and correct. Also insure that the motor armature, field and other options are properly connected.

3. Confirm that the following jumpers (JP) and switch positions (SW) are in accordance with the following requirements; also confirm that all other jumpers are in accordance with the jumper selection chart supplied with the drive.

3.1 Control Card: (See Figure 13)

SW9-1 MAN REF RANGE - Analog Master Speed Ref. Scaling
Opened Position: 5-8V maximum
Closed Position: 8-15V maximum

SW9-2 TCH/CMF - Speed/Voltage Regulator Select
Closed for speed regulator using tachometer feedback
Opened for CEMF regulator

SW9-3 DC/AC - DC/AC Tachometer Select
Closed for DC Tach feedback
Opened for AC Tach feedback

3.2 Process Interface Card (See Appendix II)

NOTE: SW position numbers in Table I, Appendix II Instructions 36A349267AP, are shown in the "NOT OPEN" position; Tach voltages are at maximum speed. Position 7 should be "CLOSED" for AC tach, "OPEN" for DC tach.

4. Verify that the DC motor has been properly installed and its shaft is free to rotate.

5. Apply power to the drive.

6. If the diagnostic LEDs on the Control card are not blinking one at a time from right to left, press and release the RESET switch on the Control card.

   If the LEDs do not begin blinking, refer to fault codes and recommended action as provided in the Troubleshooting section.

7. Check for proper direction of rotation of blower motor(s) on the DC motor(s).

8. Check the field for proper voltage. Since the drive is in field economy until a "START" is called for, the field voltage will be approximately 105 VDC for a 230 VAC input and 210 VDC for 460 VAC input.

9. Set the external speed reference potentiometer to zero and press the "START" pushbutton. The MD contactor should pick up. The diagnostic LEDs should begin blinking two at a time from right to left. The field voltage should increase to 300 VDC for 460 VAC input or 150 VDC for 230 VAC input.

10. Insure that all "STOP" buttons are functional, dropping out MD. Press the "START" button again.

11. Slowly increase the speed reference until the motor starts to rotate. If motor does not rotate, check the diagnostic LEDs. If DC Tachometer Feedback is provided in the system, the most probable cause is an incorrect tach polarity and the LEDs illuminate as follows:

    000101
    where 0 represents LED OFF
    and 1 represents LED ON

    Stop the drive. Remove power, interchange the tach leads and repeat.

12. Increase the speed reference potentiometer CW and check the motor rotation. If incorrect, remove power and interchange the motor field leads. If a DC tach is provided in the system, interchange the tachometer leads at this time.

    Most of the pots and jumpers are used in the interface circuits for the microprocessor software control circuit. Some of the jumpers are only for manufacturer's use (See Table IV for complete listing and description of these jumpers, pots and switches.) The test data sheet (Figure 18) shipped with each controller indicates the position of these adjustments as they were set in the factory. Adjustments not set in the factory can easily be made by referring to the information contained in Table IV.

NOTE: Only high impedance digital voltmeters should be used to make indicated measurements.

13. Restart the drive and slowly turn the speed-pot to mid-position (50% speed). Check actual speed by measuring tach output voltage or use a hand tach. Adjust the MAN REF SCALE potentiometer (P1) to obtain 2 VDC at SRS on the Control card. (Measure with a volt-ohmmeter between the appropriate test post and ACOM.)

14. Gradually apply maximum reference: adjust "MAN REF SCALE" potentiometer (P1) on the Control card until 4 volts is read at post SRS on the Control card. If required, change position SW9-1 (right most switch in the DIP switch package) on the Control card to change full scale reference range).

CAUTION

CONTROLLER MUST BE STOPPED WHEN CHANGING SWITCH OR JUMPER POSITIONS.

15. Confirm good motor commutation (absence of sparking) under load.

16. Slowly decrease speed. Verify stability through the full speed range. If the drive is unstable, adjust the SPDPG and SPDIG pots (P6 or P16 and P7 or P17) located on the Auxiliary Adjustment card to achieve stability. If P6 or P16 and P7 and P17 do not stabilize the drive, return them to their original position and adjust ILLPG (P4 or P14) located on the Auxiliary Adjustment card.

17. After the drive is stable throughout the full speed range, confirm that armature current does not exceed nameplate rating.

18. For motors rated 16 amps or less, jumpers JP36 and JP37 are both in the 1-2 positions or for motors rated above 16 amps both jumpers are in 2-3 position.

19. To set up the proper current feedback scaling, P13 or P18 (HP SEL or CFBSF) must be adjusted. Adjust the potentiometer so that test post HS or CS on the Auxiliary Adjustment card is the proper voltage with respect to test point ACOM as described in the following:

HORSEPOWER				JP36 JP37 POSITION	MOTOR RATED CURRENT	VOLTAGE AT HS or CS POINT
230VAC	460VAC	575VAC 630VDC	575VAC 700VDC			
125 - 150	300	—	400	2-3	440-525	A
200 - 250	400 - 800	400-1000	500-100	2-3	600-1285	B

A = 5.714 [1 - (80/RATED)]

B = 5.714 [1 - (128/RATED)]

Example:
A 500 HP motor (460 VAC) with rated current of 830 amps requires a voltage 'B' of 5.714 [1 - 128/830] or 4.83 volts at post HS (or CS) test position.

20. The acceleration and deceleration rates can be adjusted by use of the ACCEL TIME and DECEL TIME potentiometers (P3 or P13 and P2 or P12) located on the Auxiliary Adjustment card. They should be readjusted, if necessary, at this time.

21. Jog speed can be set by use of the JOG SPD potentiometer (P20) located on the Control card. Jog speed should be readjusted, if necessary at this time.

22. If the motor is unstable or erratic adjust potentiometers P6 or P16 (SPDPG) and P7 or P17 (SPDIG) until drive is stable. The system gain is related to SPDPG while the time constant is related to the ratio of SPDPG to SPDIG.

23. **The controller is now ready for operation.**

KEY POINTS

CONTROL CARD

VFB = ±4V at 100% Armature Volts
SRS = ±4V at 100% Speed Reference

+15V Power Supply = +15VDC (±5%)
-15V Power Supply = -15VDC (±5%)
+5V Power Supply = +5VDC (±5%)

PROCESS INTERFACE CARD (option)

SFB = ±4V at 100% Speed

NOTE: These voltages should be measured using high impedance meters (20K ohms/volt or higher).

HARDWARE JUMPER AND POT ADJUSTMENT

Most of the jumper selections and pot adjustments have been factory selected and adjusted. **When one of the cards has to be changed make sure the replacement card switches, jumpers and pots are set to be the same as on the replaced card.**

Hardware jumpers and pots are located on the Control card, Auxiliary Adjustment card and optional Process Interface card.

NOTE: Only a high impedance digital voltmeter should be used to make indicated measurements.

CUSTOM HARDWARE DESCRIPTION
IB# XYZ000IB0101
SHEET 1

(For details, see core controller instruction book)

CARD	JP/SW	POSITION	FUNCTION
CONTROL	SW1	OPEN	8-15V MAN SPEED REF
CONTROL	SW2	OPEN	CEMF REGULATOR ENABLED
CONTROL	SW3	OPEN	AC TACH
CONTROL	SW4	OPEN	ONE SHOT SCR TEST
CONTROL	SW5	CLOSE	4-QUADRANT OPERATION
CONTROL	SW6	CLOSE	REVERSING ENABLED
CONTROL	SW7	OPEN	STALL PROTECTION DISABLED
CONTROL	SW8	OPEN	NORMAL REGULATOR
CONTROL	JP3	1-2	MANUFACTURER'S USE ONLY
CONTROL	JP4	1-2	MANUFACTURER'S USE ONLY
CONTROL	JP5	1-2	MANUFACTURER'S USE ONLY
CONTROL	JP6	1-2	MANUFACTURER'S USE ONLY
CONTROL	JP7	1-2	MANUFACTURER'S USE ONLY
CONTROL	JP8	3-4	MANUFACTURER'S USE ONLY
CONTROL	JP10	1-2	MANUFACTURER'S USE ONLY
CONTROL	JP11	1-2	MANUFACTURER'S USE ONLY
CONTROL	JP12	1-2	MANUFACTURER'S USE ONLY
CONTROL	JP13	1-2	MANUFACTURER'S USE ONLY
CONTROL	JP1	1-2	DRIVE CURRENT RATING
CONTROL	JP2	1-2	DRIVE CURRENT RATING
CONTROL	JP9	1-2	440-480 VAC INPUT
	CNTL1-CNTL2	IN	DRIVE CONTROL JUMPER
	XSTP-REF24	IN	AUXILIARY STOP JUMPER
	INIT-REF24	IN	INITIATE TRIP JUMPER

EXAMPLE

FIGURE 18
TYPICAL DATA SHEET

TABLE IV

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Control Card	Switch	SW9-1	MAN REF RANGE	Used in conjunction with P1 (MAN REF SCALE) to scale MSR input. Open for reference voltage 5—8V maximum. Close for reference voltage 8—15V maximum.
		SW9-2	TCH/CMF	Used to select regulating mode. Close for speed regulator (tach feedback - requires Process Interface Kit Option). Open for CEMF regulation.
		SW9-3	DC/AC	Used to select tachometer type: Close for DC tachometer feedback. Open for AC tachometer feedback.
		SW9-4	SCR TEST	Used to perform SCR cell tests. To request SCR test one time only, close SW9-4, press RESET, open SW9-4 and press START. The test will be run once and the drive will stop.
		SW9-5	REG/NON	Used to select between one or four quadrant operation: Close for regenerative operation. Open for non-regenerative operation.
		SW9-6	REV/NON	Used to allow reversal of regenerative controllers: Close to allow reversal (requires a regenerative controller). Open to disable reversal.
		SW9-7	STALL TRIP ENABLE	Used to enable Stall Trip circuit: Close to enable. Open to disable.
		SW9-8	ILOOP	Used to select inner/outer regulator loop type: Close to run inner current regulator without outer speed regulator. Open for normal outer speed or CEMF regulator with inner current loop.
		SW2	RESET	This switch may be momentarily pushed in order to reset the microprocessor and its associated I/O expander chip.

NOTE: After changing any switch settings the reset switch must be toggled.

NOTE: To "open" a switch position, depress the side of the switch labeled "OPEN" or "OFF". To close a switch position, depress the side of the switch labeled with a number or "ON".

TABLE IV

LOCATION	JUMPER OR POT	NUMBER	NOMEN CLATURE	DESCRIPTION
Control Card	Pot	P1	MAN REF SCALE	Used to scale MSR input. This pot provides scaling by a factor as determined by SW9-1 status: SW9-1 open: 0.44 to 0.82 SW9-1 closed: 0.26 to 0.67
		P2	IMET SCALE	Used to adjust the output to the load voltmeter circuit. 8 volt output equals 100% of rated current.
		P4	OFFSET	Adjusted by manufacturer not to be adjusted in the field
		P19	ILIM	Used to scale the CLA input.
		P20	JOG SPD	Used to scale the JOGR input.
	Jumper	JP2 JP5 JP38		1-2: Should be in this position for proper operation. 2-3: Only for manufacturer use.
		JP19 JP20		Always connect as follows: JP-19: 1-2 JP-20: 3-4
		JP39		Phase Loss Detector Jumper 1-2: for 440 to 480 VAC (and 575 VAC). 2-3: for 220 to 240 VAC.
		JP21 JP22 JP23 JP24		Program memory configuration jumpers. Connect 1-2
		JP36 JP37		Scaling for IL1 and IL3. Set 1—2 for motors rated $<=$ 16 amps. Set 2—3 for motors rated $>$ 16 amps.
		JP40		Field Loss enable Connect 1—2 for field loss detection and trip. Connect 2—3 to disable this function.
	Switch	SW1-1		Open: Should be in this position for proper operation. Closed: Only for manufacturers use.

TABLE IV (continued)

LOCATION	JUMPER OR POT	NUMBER	NOMEN-CLATURE	DESCRIPTION
Auxiliary Adjustment Card	Pot	P1 or P11	MIN SPD	Used to adjust controller minimum speed. Adjusts minimum speed from zero to approximately full speed.
		P2 or P12	DECEL TIME	Used to adjust the linear deceleration time (between zero and approximately 300 seconds), forward or reverse.
		P3 or P13	ACCEL TIME	Used to adjust the linear acceleration time (between zero and approximately 300 seconds), forward or reverse.
		P4 or P14	ILPPG	Used to adjust the current loop integral and proportional gains. Also sets the current loop discontinuous gain to twice the proportional gain.
		P5 or P15	COMP	Used to adjust for positive or negative IR compensation. 0 COMP at mid position + COMP CW; — COMP CCW
		P6 or P16	SPDPG	Used to adjust the proportional gain of the CEMF or outer speed loop.
		P7 or P17	SPDIG	Used to adjust the integral gain of the CEMF or outer speed loop.
		P13 or P18	CFBSF or HPSEL	Used to adjust the current feedback scale factor (and output voltage on test post HS or CS). Refer to the Start-up Procedure, item 19 for HS or CS adjust procedure.

TEST POSTS

As a further aid to troubleshooting the DC-200 drive controller, a series of test posts have been provided to allow for specific signal observation and/or measurements. Test posts not defined in the following listing are for manufacturer's use only and should not have instrumentation connected to them.

CONTROL CARD TEST POSTS

TEST POST NAME	DESCRIPTION	NOMINAL VALUE
ACOM	Common for analog circuit power supplies.	
DCOM	Common for digital circuit power supply.	
-15V	-15 VDC	-15 volts ±5%
+15V	+15 VDC	+15 volts ±5%
+5V	+5 VDC	+5 volts ±5%
SRS	Scaled speed reference signal.	4 volts at top speed
VFB	Armature voltage feedback signal. Positive when forward motoring. Maximum external loading is 1 mA.	4 volts at rated armature voltage

AUXILIARY ADJUSTMENT TEST POSTS

TEST POST NAME	DESCRIPTION
HS or CS	Output voltage of CFBSF potentiometer. Used for current feedback scaling.
COM	Reference common for test post HS or CS, also, analog circuit common.

TROUBLESHOOTING

The DC-200 drive controller detects and displays most of the faults that could occur. It displays the faults with a series (interpreted as a binary number) of LEDs on the Control card or with a (name and number) fault message on the optional Digital Operator Station.

The following table lists predefined fault codes in order by the assigned fault number. LEDs are assigned a "bit weight" (based on a power of two) from right to left according to whether the LED is "ON" or "OFF". If any LED is "BLINKING", add 64 to the total value to obtain the fault number. For example.

	BINARY NUMBER(s)						
LED INDICATION	**32**	**16**	**8**	**4**	**2**	**1**	**FAULT NUMBER**
Steady Display	0	1	1	1	0	0	16 + 8 + 4 = 28
Blinking Display	0	1	0	1	0	0	16 + 4 + 64 = 84

NOTE: 0 = LED OFF 1 = LED ON 1 = Blinking LED

NOTE: Blinking lights in succession from right to left are normal operating system indicators. One light blinking from right to left indicates "Ready" to run. Two adjacent lights blinking in pairs from right to left indicate "Running".

TROUBLESHOOTING GUIDE FOR THE DC-200

It is possible that the Control card or the Digital Operator Station may not show a fault but in reality a fault does exist. Before replacing anything the following checks are recommended.

1. Is power applied to the drive and is it the correct voltage?

2. Check the following and replace any open fuses (see fuse location and nomenclature on the information label inside the controller door).
   FU1, 2, 3 (Power Fuses)
   FU4 (Regenerative Controller only —DC Fuse)*
   CFU1, 2, 3, 4 (High Voltage Fuses)
   CFU6, 7, 8 (Low Voltage Fuses)

3. If the display still does not illuminate, the Control card or Operator Station should be replaced. If the display does illuminate proceed to the appropriate fault display in the following listing.

The faults listed in the table below may be divided into four categories:

L — The fault is latched, i.e., once it occurs it will not clear itself even if the cause of the fault no longer exists. The controller must be reset to clear it.

N — The fault is non-latched, i.e., it will clear itself if the cause of the fault no longer exists.

T — The fault trips the drive, i.e., it causes the controller to remove power from the motor. The controller must be reset. All trip faults are latched and are thus designated (L,T).

A — The fault is annunciated only, i.e., if the drive is running it will not cause the drive to shut down. However, an annunciated fault will prevent the drive from starting if it is not running. Annunciated faults in general are non-latched and are thus designated (N,A).

There are several ways to reset the controller. A software reset may be executed either from the Digital Operator Station, or by serial command from a device connected to the serial link. A hardware reset may be executed by pressing the reset switch on the Control card, or by momentarily interrupting incoming power to the drive. (see also RSET description for connector 6PL). Most latched faults may be adequately cleared by a software reset. However, some faults should be cleared by a hardware reset. These are indicated by the letter "H". If a software reset does not seem to restore the drive to normal operation, a hardware reset should be attempted.

*NOTE: On Regenerative Controllers with paralleled DC fuses, replace both fuses, even if only one fuse is blown.

SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT*	FAULT NUMBER	FAULT NAME	DESCRIPTION/PROBABLE CAUSES/REMEDIES
000001	FL01	IOC	Instantaneous DC overcurrent (L,T) • Mechanical problem. • SCR Misfiring — check test post CFB with an oscilloscope. • Current regulator response too high — reduce gains, by adjusting ILPIG. • Current feedback isolator circuit failure — check 1CPL.
000010	FL02	OVRSP	Overspeed (sensed by analog tach circuitry) (L,T) • Tach feedback circuit failure — check 7PL-3 and tach wiring. • Speed regulator response too high — reduce gain, by adjusting SPDPG & SPDIG. • Confirm that at maximum speed reference, 4 VDC appears at SRS, and readjust MAN REF SCALE (P1) as necessary. • Confirm that at maximum speed, 4 VDC appears appears at SFBB and readjust TACH SCALE (P1) as necessary.
000011	FL03	Spare	
000100	FL04	FLOSS	Loss of field current (L,T) • Open fuse in field circuit. • Field feedback circuit failure—check connection AF or DZ on the motor field power module. • Loose connection in field power circuit.
000101	FL05	TPOL	Tach polarity incorrect (L,T) • Reverse tach connections.
000110	FL06	Spare	
000111	FL07	RESET	Reset while running (L,T) • Drive powered up with RUN or JOG closed — Check 6PL-12(3TB58) and 13(3TB57) for -24 VDC. • Momentary loss of incoming power while running.
001000	FL08	TOC	Timed overcurrent (L,T) • Drive overloaded — check application. • Check field current.
001001	FL09	LNKTR	Serial link trip (L,T) • Drive told to trip by serial link — check for external fault.
001010	FL10	LNKFL	Serial link fault (L,T) • Serial link communications locked up. • Check clear to send handshakes at 12PL-7 and 14.
001011	FL11	Spare	
001100	FL12	Spare	

NOTES:
*Consists of 6 LEDs;
0 = LED OFF, 1 = LED ON, 1 = LED BLINKING

(L,T) indicates Latched fault which Trips the drive; which requires a reset to clear the fault.

(N,A) indicates Non-latched fault, which is Annunciated only; does not shut down drive, but prevents drive from restarting.

SPECIFIC FAULT TROUBLESHOOTING

DIAGNOSTIC READOUT*	FAULT		DESCRIPTION/PROBABLE CAUSES/REMEDIES
	NUMBER	NAME	
001101	FL13	Spare	
001110	FL14	ILOOP	Current loop regulator saturated (L,T) • Main contactor not picking up — check pilot relay operation. • Loss of current feedback — 1CPL (3TB MUP3). • Armature not connected. • DC fuse open. • 5PL disconnected.
001111	FL15	VFBLS	Loss of armature voltage feedback (L,T) • Motor rotating while no armature volts applied. SFB circuit failure — check for SFB equal to zero on the Process Interface card. • VFB circuit failure — check test point VFB on the Control card.
010000	FL16	Spare	
010001	FL17	XSTOP	Coast stop trip (L,T) • Check external input at 6PL-16 (3TB1).
010010	FL18	STALL	Stalled overcurrent (L,T) • Check for mechanical problems. • Check settings for stall gain and threshold. • Check field current.
010011	FL19	OVERV	Excessive armature voltage (L,T) • Speed or CEMF regulator settings unstable. • Armature voltage feedback circuit failure — check test point VFB.
010100	FL20	TLOSS	Loss of Tachometer feedback (L,T) • Check external tach connections. • Jumpers or tach switch positions incorrect on the Process Interface card. • Tach feedback circuit failure—check 7PL-3 (test point SFB) and 13 (test point TPN) for tach voltage.
010101	FL21	NCTRP	Trip from N.C. input on 6PL-12 (RUN) • Check connection to 6PL-12 and/or 3TB point 58. • Check external input at 6PL-12.
010110	FL22	PHLOS	AC phase loss (L,T) • Fuse blown in AC line. • Low AC line or momentary line notch.
010111	FL23	Spare	
011000	FL24	U3 FL	Bad RAM on U3 8155 (L,T) • Replace the Control card.

NOTES:

*Consists of 6 LEDs:
0 = LED OFF, 1 = LED ON, 1 = LED BLINKING

(L,T) indicates Latched fault which Trips the drive; which requires a reset to clear the fault.

(N,A) indictes Non-latched fault, which is Annunciated only; does not shut down drive, but prevents drive from restarting.

SPECIFIC FAULT TROUBLESHOOTING *(continued)*

DIAGNOSTIC READOUT*	FAULT		DESCRIPTION/PROBABLE CAUSES/REMEDIES
	NUMBER	NAME	
011001	FL25	COMTR	SCR commutation failure (L,T) • Low AC line. • HP SEL and JP36 & 37 adjustments gain too high. • Incorrect connection of AC line CTs.
011010	FL26	ACIOC	Circulating AC line fault (L,T) • Bad SCR. • Low or noisy AC line. • Incorrect setting of either the HP SEL on the Auxiliary Adjustment card, or JP36 & 37 on the Control card.
011011	FL27	LSYNC	Loss of AC line sync signal (L,T) • Check FL22 causes first. • Check for 50/60 Hz at TP SY and U4-12 on the Control card. • AC line frequency too high or too low.
011100	FL28	Spare	
011101	FL29	Spare	
011110	FL30	Spare	
011111	FL31	NOSEQ	Phase sequence indeterminate (L,T) • Noisy or bad AC line — check line feedback circuits.
100000	FL32	TOCAN	Timed overcurrent annunciation (N,A) • Drive overloaded — check application.
100001	FL33	Spare	
100010	FL34	PLLSY	AC line Phase Locked Loop out of sync (N,A) • Noisy AC line.
100011	FL35	OSPD	Motor rotating before run commanded (N,A) • Motor driven externally (overhauling load on coast-down situation). • Adjust SFB ZERO (P7) on Process card to obtain 0 volts at SFB with 0 speed (requires Process Interface card option).
100100	FL36	LNKER	Bad message detected on serial link by drive or other unit (N,A) • Check for excessive noise on link.
100101	FL37	Spare	
100110	FL38	Spare	
100111	FL39	Spare	

NOTES:
*Consists of 6 LEDs
0 = LED OFF, 1 = LED ON, 1 = LED BLINKING

(L,T) indicates Latched fault which Trips the drive; which requires a reset to clear the fault.

(N,A) indicates Non-latched fault, which is Annunciated only; does not shut down drive, but prevents drive from restarting.

SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT*	FAULT		DESCRIPTION/PROBABLE CAUSES/REMEDIES
	NUMBER	NAME	
101000	FL40	MXSPD	Speed reference greater than maximum speed adjustment (N,A). • Check scaling of reference and MAN REF SCALE pot • (P1) on the Control card.
101001	FL41	KEYOF	Communications Link Error. • Consult factory
101010	FL42	RLOGC	Run logic error (N,A) • Check RUN, JOG, and XSTP inputs for mismatch.
101011	FL43	Spare	
101100	FL44	Spare	
101101	FL45	SLOOP	Speed loop tracking fault (L,T) • Drive entered current limit or CEMF limit when tracking enabled. • Speed loop error junction cannot be brought into regulation.
101110	FL46	Spare	
101111	FL47	Spare	
110000	FL48	Spare	
110001	FL49	Spare	
110010	FL50	U4BAD	U4RAM, timer or ALU is bad (L,T) • Replace the Control card.
110011	FL51	U1ROM	U1 ROM is bad (L,T) • Replace the Control card.
110100	FL52	Spare	
110101	FL53	Spare	
110110	FL54	Spare	
110111	FL55	Spare	
111000	FL56	Spare	
111001	FL57	Spare	

NOTES:
*Consists of 6 LEDs:
0 = LED OFF, 1 = LED ON, 1 = LED BINKING

(L,T) indicates Latched fault which Trips the drive; which requires a reset to clear the fault.

(N,A) indicates Non-latched fault, which is Annunciated only; does not shut down drive, but prevents drive from restarting.

SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT*	FAULT		DESCRIPTION/PROBABLE CAUSES/REMEDIES
	NUMBER	NAME	
111010	FL58	Spare	
111011	FL59	Spare	
111100	FL60	Spare	
111101	FL61	Spare	
111110	FL62	Spare	
111111	FL63	Spare	
000000	FL64	Spare	
000001	FL65	A1FOP	SCR A1F is not operating (L,T) • Check for open gate lead/replace SCR A1F.
000010	FL66	A2FOP	SCR A2F if not operating (L,T) • Check for open gate lead/replace SCR A2F.
000011	FL67	A3FOP	SCR A3F is not operating (L,T) • Check for open gate lead/replace SCR A3F.
000100	FL68	A4FOP	SCR A4F is not operating (L,T) • Check for open gate lead/replace SCR A4F.
000101	FL69	A5FOP	SCR A5F is not operating (L,T) • Check for open gate lead/replace SCR A5F.
000110	FL70	A6FOP	SCR A6F is not operating (L,T) • Check for open gate lead/replace SCR A6F.
000111	FL71	FWDOP	All forward SCRs are not operating (L,T) • Confirm gate signals present. • Check 5PL.
001000	FL72	ARMOP	High armature voltage detected during SCR test (L,T) • Check for open armature circuit or blown DC fuse. • Check 5PL.
001001	FL73	A1ROP	SCR A1R is not operating (L,T) • Check for open gate lead/replace SCR A1R.

NOTES:
*Consists of LEDs:
0 = LED OFF, 1 = LED ON, 1 = LED BLINKING

(L,T) indicates Latched fault which Trips the drive; which requires a reset to clear the fault.

(N,A) indicates Non-latched fault, which is Annunciated only; does not shut down drive, but prevents drive from restarting.

SPECIFIC FAULT TROUBLESHOOTING (continued)

DIAGNOSTIC READOUT*	FAULT		DESCRIPTION/PROBABLE CAUSES/REMEDIES
	NUMBER	NAME	
001010	FL74	A2ROP	SCR A2R is not operating (L,T) • Check for open gate lead/replace SCR A2R.
001011	FL75	A3ROP	SCR A3R is not operating (L,T) • Check for open gate lead/replace SCR A3R.
001100	FL76	A4ROP	SCR A4R is not operating (L,T) • Check for open gate lead/replace SCR A4R.
001101	FL77	A5ROP	SCR A5R is not operating (L,T) • Check for open gate lead/replace SCR A5R.
001110	FL78	A6ROP	SCR A6R is not operating (L,T) • Check for open gate lead/replace SCR A6R.
001111	FL79	REVOP	All reverse Power Modules are not operating (L,T) • Confirm gate signals present. • Check 5PL.
010000	FL80	SCTFL	SCR short circuit test failed (L,T) • Check for multiple power module failures.
010001	FL81	A1FSH	SCR A1F or A4R is shorted (L,T) • Check/replace SCRs A1F and A4R.
010010	FL82	A2FSH	SCR A2F or A5R is shorted (L,T) • Check/replace SCRs A2F and A5R.
010011	FL83	A3FSH	SCR A3F or A6R is shorted (L,T). • Check/replace SCRs A3F and A6R.
010100	FL84	A4FSH	SCR A4F or A1R is shorted (L,T) • Check/replace SCR A4F and A1R.
010101	FL85	A5FSH	SCR A5F or A2R is shorted (L,T) • Check/replace SCRs A5F and A2R.
010110	FL86	A6FSH	SCR A6F or A3R is shorted (L,T) • Check/replace SCRs A6F and A3R.

NOTES:

*Consists of 6 LEDs:
0 = LED OFF, 1 = LED ON, 1 = LED BLINKING

(L,T) indicates Latched fault which Trips the drive; which requires a reset to clear the fault.

(N,A) indicates Non-latched fault, which is Annunciated only; does not shut down drive, but prevents drive from restarting.

SPECIFIC FAULT TROUBLESHOOTING *(continued)*

DIAGNOSTIC* READOUT	FAULT		DESCRIPTION/PROBABLE CAUSES/REMEDIES
	NUMBER	NAME	
010111	FL87	Spare	
011000	FL88	Spare	
011001	FL89	Spare	
011010	FL90	Spare	
011011	FL91	Spare	
011100	FL92	Spare	
011101	FL93	Spare	
011110	FL94	Spare	
011111	FL95	Spare	
100000	FL96	Spare	
100001	FL97	OCTFL	SCR open circuit failed (L,T) • Check 5PL. • Check for open armature leads/fuse.
100010	FL98	OCTMF	Multiple open Power Modules (L,T) • Check for open gate leads. • Check 5PL.
100011	FL99	ALLOP	All Power Modules tested open (L,T) • Check for Power Connection card not being connected. • Check for open DC fuse or armature. • Check for contactor operation.

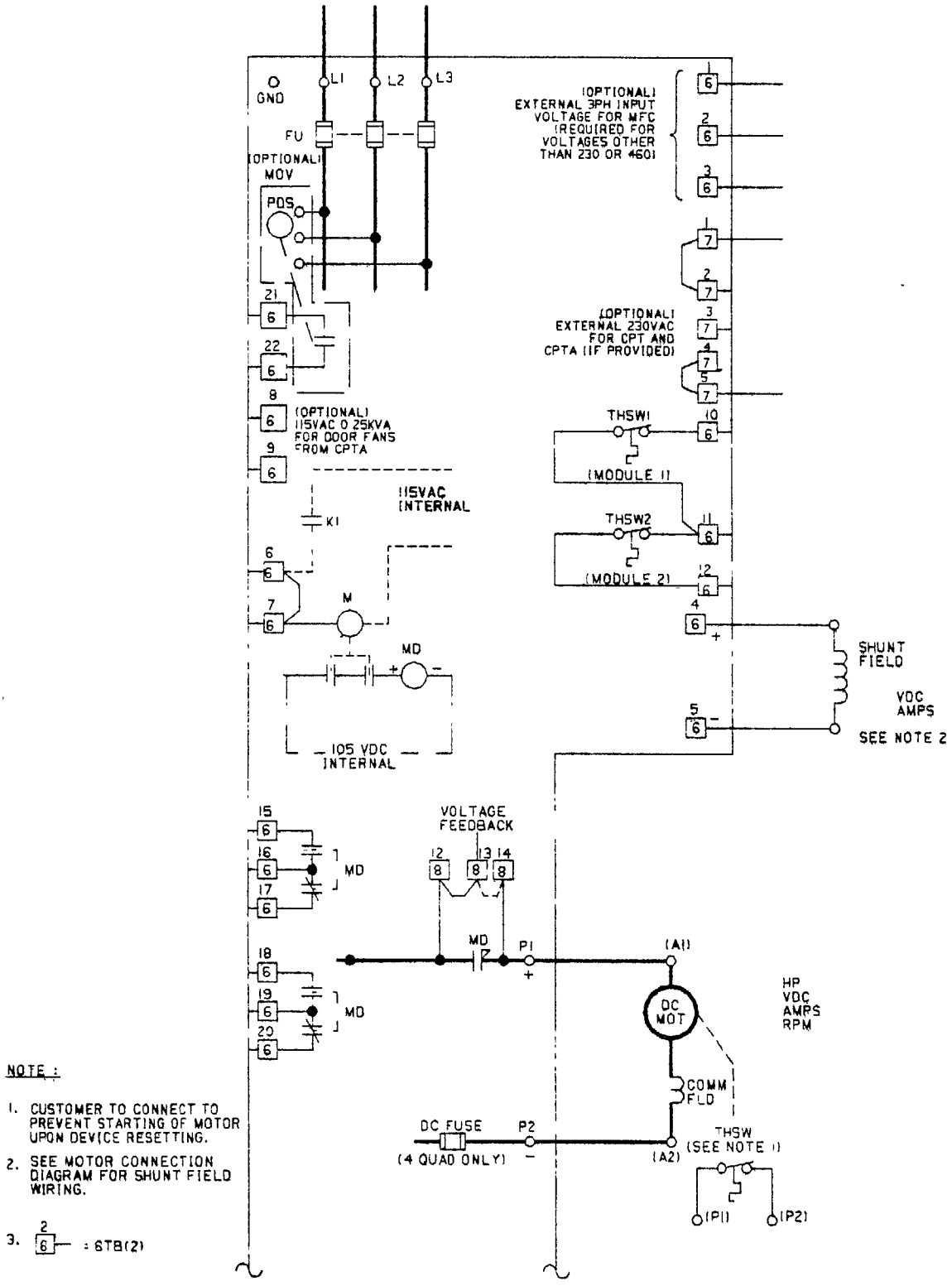
NOTES:
*Consists of 6 LEDs:
0 = LED OFF, 1 = LED ON, 1 = LED BLINKING

(L,T) indictes Latched fault which Trips the drive; which requires a reset to clear the fault.

(N,A) indicated Non-latched fault, which is Annunciated only; does not shut down drive, but prevents drive from restarting.

NOTE

Controller software provides a power bridge SCR self test. It is unlikely but not entirely impossible that the test may not provide sufficient current to verify the SCRs as functional. (Various factors affect this: design overload conditions, line voltage, motor armature resistance, inductance, DC power lead length, wire size, etc.) In the event this situation is encountered, assistance may be obtained from **GENERAL ELECTRIC COMPANY, SPEED VARIATOR PRODUCTS OPERATION, ERIE, PA. (Telephone 814-455-3219)**



NOTE :

1. CUSTOMER TO CONNECT TO PREVENT STARTING OF MOTOR UPON DEVICE RESETTING.
2. SEE MOTOR CONNECTION DIAGRAM FOR SHUNT FIELD WIRING.
3. 2/6 — : 6TB(2)

FIGURE 19
ELEMENTARY DIAGRAM — SHEET 1

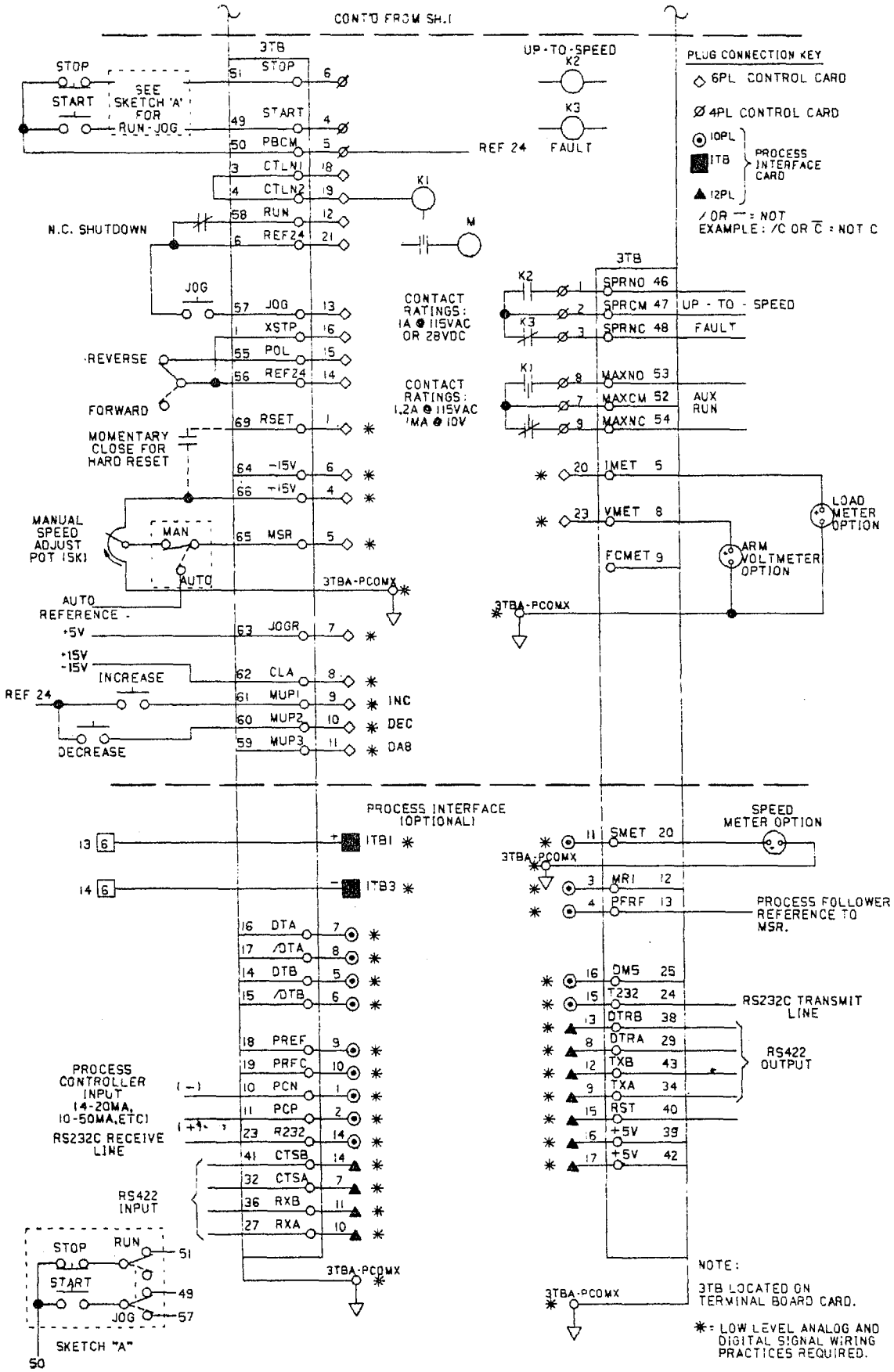
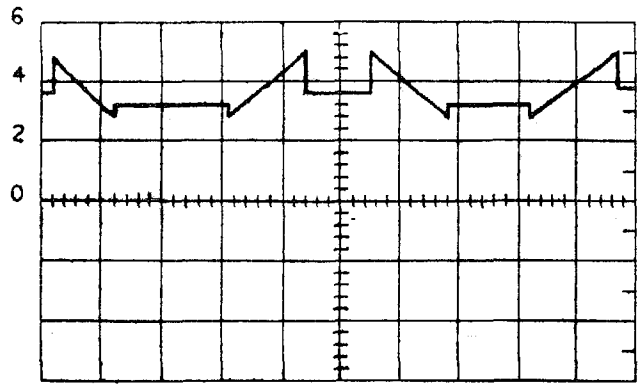
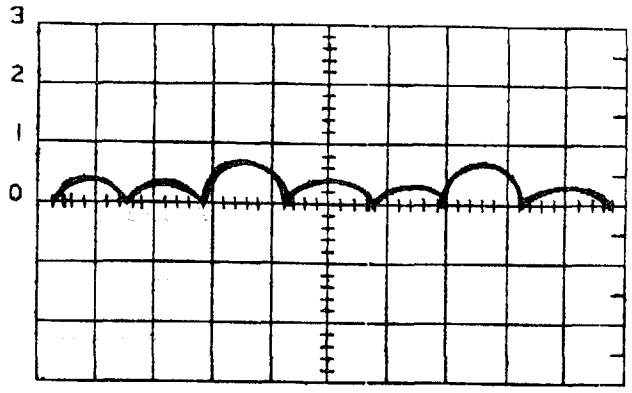


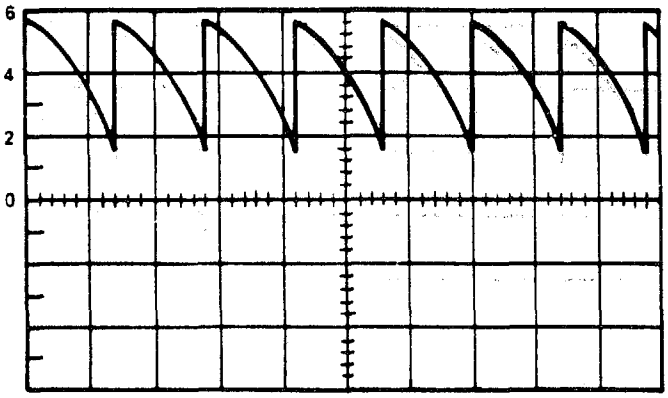
FIGURE 20
ELEMENTARY DIAGRAM — SHEET 2



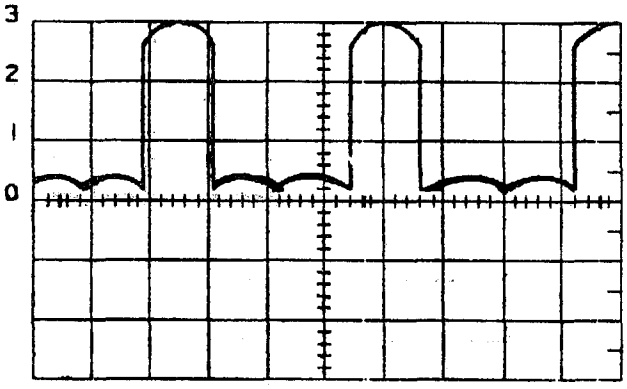
(2MS/DIVISION)
FIGURE 21
VOLTAGE FEEDBACK (VFB) AT
DISCONTINUOUS CURRENT



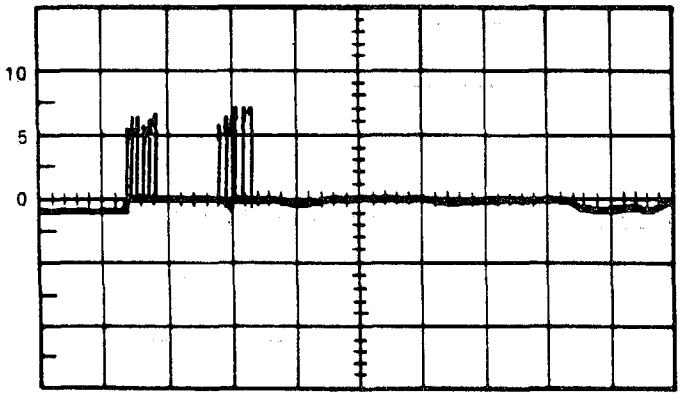
(2MS/DIVISION)
FIGURE 22
CURRENT FEEDBACK (IMET) AT
DISCONTINUOUS CURRENT



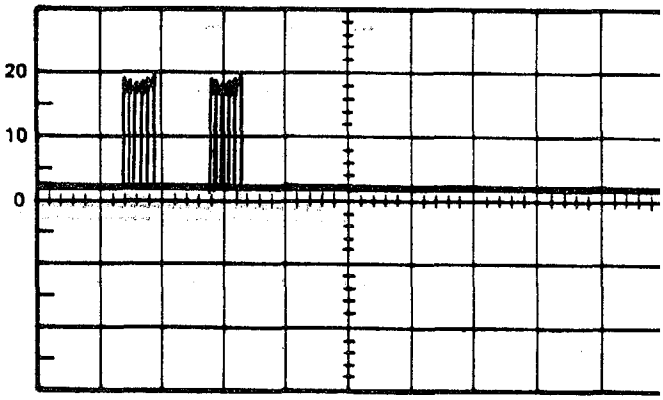
(2MS/DIVISION)
FIGURE 23
VOLTAGE FEEDBACK (VFB) AT
CONTINUOUS CURRENT



(2MS/DIVISION)
FIGURE 24
CURRENT FEEDBACK (IMET) AT
CONTINUOUS CURRENT



(2MS/DIVISION)
FIGURE 25
GATE PULSES WITH GATE CONNECTED



(2MS/DIVISION)
FIGURE 26
GATE PULSES WITH GATE DISCONNECTED

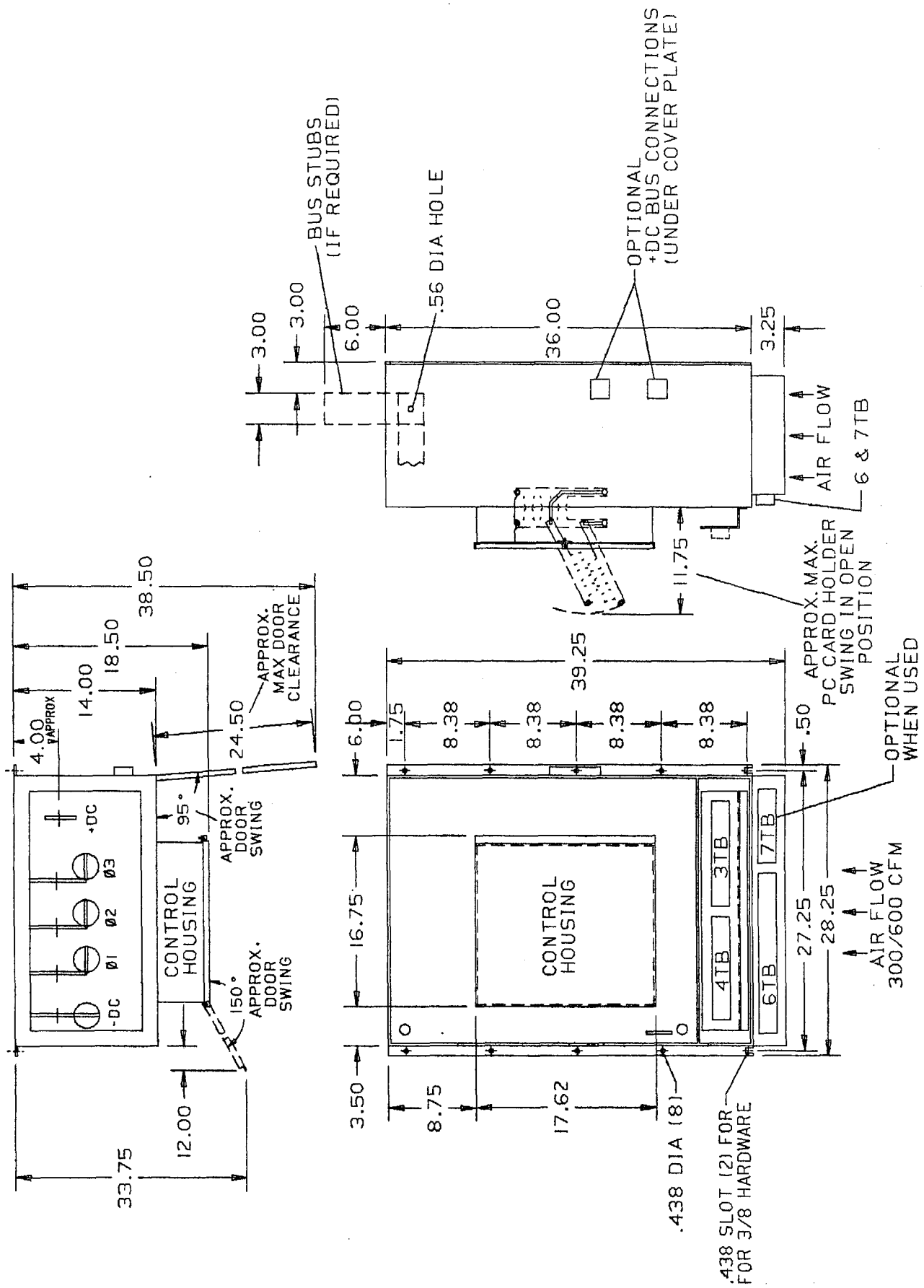


FIGURE 27
DC-200 CONTROLLER ONLY — OUTLINE

ENCLOSURE DIMENSIONS (see note)

575 VAC			
HP	D"	H"	W"
400 to 800	20	90	48
900 to 1000	20	90	60

460 VAC			
HP	D"	H"	W"
300 to 500	20	90	48
600 to 800	20	90	60

230 VAC			
HP	D"	H"	W"
150 to 200	20	90	48
250 to 300	20	90	60

NOTE: Enclosure dimension W (width) may vary from the tables shown due to optional equipment ordered.
See your Custom Instruction Book for exact dimensions of enclosures provided.

SPARE AND RENEWAL PARTS

PRINTED CIRCUIT CARDS

DESCRIPTION	CATALOG NUMBER
Control Card	531X101CCGA-M3
Power Connection Card (Regenerative)	531X121PCRA-Z1
Power Connection Card (Non-Regenerative)	531X122PCNA-Z1
Auxiliary Adjustment Card	531X143AADA-Z1
Terminal Board Card (3TB)	531X136TBGA-Z1 or 531X170TBSA-Z1
Process Interface card (option)	531X133PRUA-Z1
Digital Operators Station (option)	531X133PRUA-Z1 6VHHP1A1

SUB ASSEMBLIES AND COMPONENTS (NOT HP RELATED)

DESCRIPTION			CATALOG NUMBER
Motor Field Module			104X125DC012
Control Power Transformer (230V AC)			36B605538AFG01
Control Power Transformer (460V AC)			36B605538AFG02
Suppressor			36A358207AAG01
Fan (small)			104X215CA003
Fan (large)			104X215CA004
Current Transformer Assembly			36A358218YBG02
Fuses (Control Power)	CFU7	0.5 amps	104X109AD009
	CFU6	2 amps	104X109AD085
	CFU4	3.2 amps	104X109BC010
	CFU1-3	15 amps	104X109BC015
	CFU8	0.5 amps	104X109AD009
Fuses (Exciter Assembly)		30 amps	104X109AD084

SPARE AND RENEWAL PARTS (HP RELATED — 230 VAC)

DESCRIPTION	CATALOG NUMBER	HORSEPOWER				
		125	150	200	250	300
AC Power Fuse	104X109AG _ _ _	027	027	030	030	32
DC Power Fuse 4Q only	104X109AG _ _ _	027	028	030	032	034
Armature SCRs	104X125DA _ _ _	089	089	123	137	137
Armature SCRs (with leads)	36A353476AX _ _ _	G01	G01	G02	G03	G03
DC Contactor	104X106AE _ _ _	008	008	008	009	
DC Contactor	104X106MA _ _ _	—	—	—	—	G01
Curr. Xfmer. Asm.	36A358218YA _ _ _	G01	G01	G02	G03	G03

SPARE AND RENEWAL PARTS (HP RELATED — 460 VAC)

DESCRIPTION	CATALOG NUMBER	HORSEPOWER					
		300	400	500	600	700	800
AC Power Fuse	104X109AG _ _ _	027	030	032	032	032	034
DC Power Fuse 4Q only	104X109AJ _ _ _	025	025	025*	025*	025*	026*
Armature SCRs	104X125DA _ _ _	089	123	137	137	137	137
Armature SCRs (with leads)	36A353476AX _ _ _	G01	G02	G03	G03	G03	G03
DC Contactor	104X106AE _ _ _	008	008	009	009		
DC Contactor	104X106MA _ _ _	—	—	—	—	G01	G01
Curr. Xfmer. Asm.	36A358218YA _ _	G01	G02	G02	G02	G02	G02

SPARE AND RENEWAL PARTS (HP RELATED — 575 VAC, 630 VDC MOTORS)

DESCRIPTION	CATALOG NUMBER	HORSEPOWER						
		400	500	600	700	800	900	1000
AC Power Fuse	104X109AR ___	027	029	031	033	027*	027*	029*
DC Power Fuse 4Q only	104X109AW ___	006	007	009	006*	006*	007*	007*
Armature SCRs w/leads	36A353476AX __	G03	G03	G03	G03	G03	G03	G03
Armature SCRs	104X125DA ___	137	137	137	137	137	137	137
DC Contactor	104X106AE ___	008	008	008	008			
	104X106MA ___					G01	G01	G01

SPARE AND RENEWAL PARTS (HP RELATED — 575 VAC, 700 VDC MOTORS)

DESCRIPTION	CATALOG NUMBER	HORSEPOWER						
		400	500	600	700	800	900	1000
AC Power Fuse	104X109AR ___	027	027	029	031	032	027*	027*
Armature SCRs w/leads	36A353476AX __	G03	G03	G03	G03	G03	G03	G03
Armature SCRs	104X125DA ___	137	137	137	137	137	137	137
DC Contactor	104X106AE ___	007	008	008	008	009		
	104X106MA ___						G01	G01

* 2 per drive required

NOTE: On Regenerative Controllers with paralleled DC fuses, replace both fuses, even if only one is blown.

ISOLATION TRANSFORMER RECOMMENDATIONS

The following table provides a guide to the proper selection of isolation transformer KVA as a function of drive horsepower.

NOTE: One drive per transformer is listed if more than one drive is to be applied to a single transformer, contact **GENERAL ELECTRIC COMPANY, SPEED VARIATOR PRODUCTS OPERATION, ERIE, PA. TELEPHONE (814-455-3219)** for assistance.

HP	TRANSFORMER KVA
100	118
125	145
150	175
200	220
250	275
300	330
400	440
500	550
600	660
700	750
800	1000
900	1000
1000	Consult factory

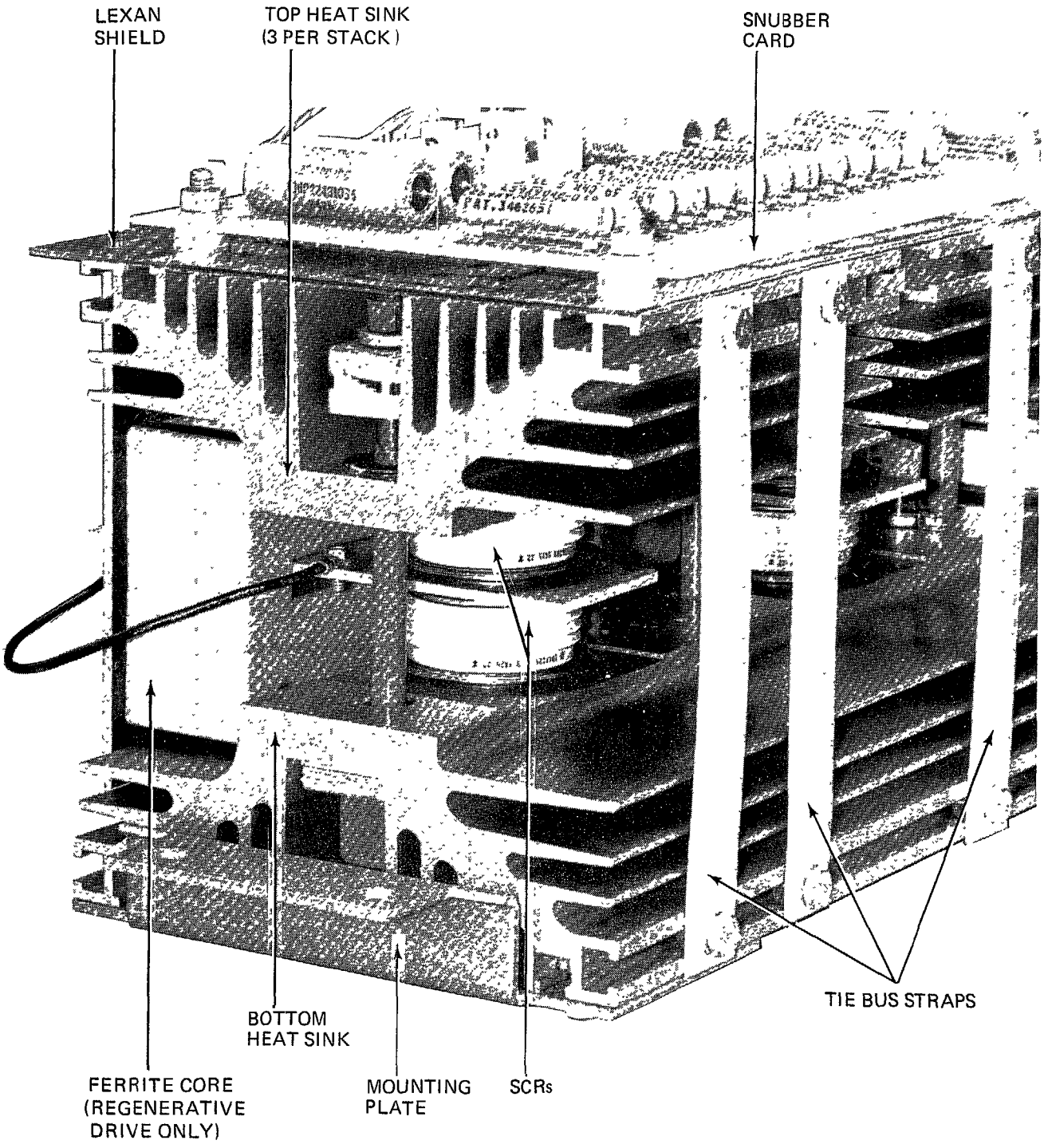


FIGURE 28
DC-200 SCR STACK ASSEMBLY

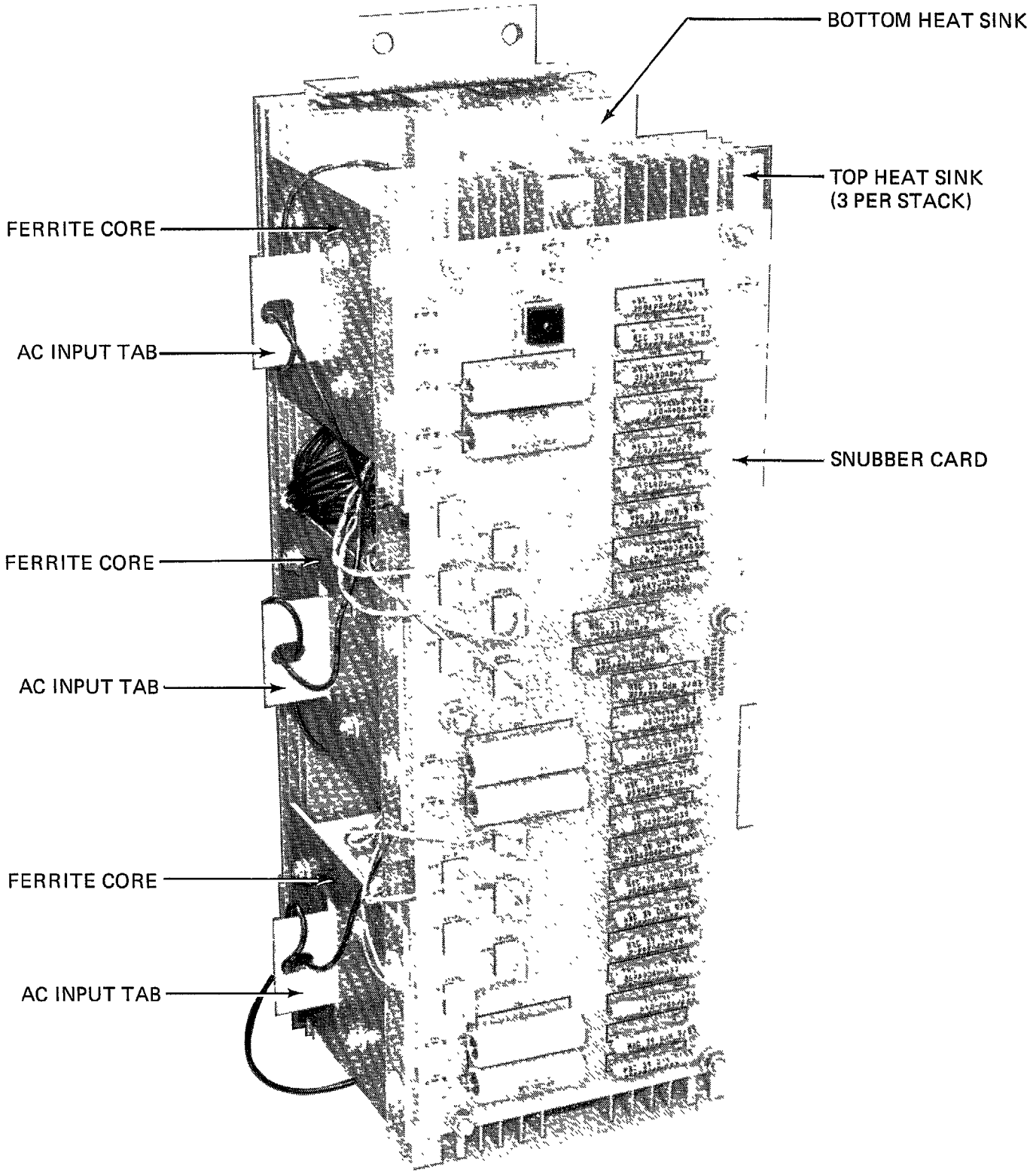


FIGURE 29
DC-200 SCR ASSEMBLY

APPENDIX I
SCR/STACK ASSEMBLY REMOVAL/REPAIR/REPLACEMENT

The following procedure is suggested when it becomes necessary to replace a defective SCR.

Right Hand SCR Stack Removal Procedure

- o Remove all power from the equipment.
- o Remove the SCR firing circuit plugs from the snubber card for the SCRs located in the right hand module only. It is not necessary to remove the SCR firing circuit plugs for the left hand module.
- o Remove the AC wires (1AC__, 2AC__, 3AC__) from the snubber card.
- o Remove the 6 mounting nuts which hold the snubber card in place being careful not to let the bolts drop into the bottom of the controller.
- o Remove the harness retainers from their captive position.
- o Carefully roll the snubber card off the stack and over onto the left hand stack.
- o Remove the three AC line input bolts, again being careful not to lose any hardware.
- o Remove the electrical connection (bus) at the bottom of the stack.
- o Likewise remove the stack mounting bolts (qty 4) at the top and bottom of the stack.
- o The stack assembly can now be lifted off the panel and placed on a workbench for further disassembly.

Left Hand SCR Removal Procedure

- o Follow the same procedure outlined above (except removal of the snubber card), i.e., removal of appropriate SCR firing circuit plugs, 4AC__, 5AC__, 6AC__ leads from the snubber card, mounting hardware, electrical connections.

SCR Replacement Procedure

- o Identify the name of the defective SCR (from fault codes) and SCR location (See Figures 31 and 32).
- o Place the stack assembly on its side and disassemble the section containing the defective SCR.
- o Alternately loosen each nut on the clamp assembly to keep even pressure on the good cell.
- o As each part is removed from the stack assembly, place it on a clean surface and in the same orientation and order in which they were removed.
- o Remove the defective cell. Clean and lubricate the heatsinks before reassembling. Place a few drops of GE SF1153 silicone oil in the center of both heatsink mounting surfaces. Polish the interface area until clean, using a combination of back and forth, sideways, and circular motions with a lint-free cloth or tissue. Finally place a few drops of GE SF1153 silicone oil on the mating surfaces of the heatsink and use a lint-free cloth or tissue to spread the oil evenly. Wipe off excess oil.
- o Clean and lubricate the SCR cell. Place the cell on a flat table and clean the anode and cathode faces with GE SF1153 oil as described above. Apply a thin film of silicone oil to cover the mating surfaces and wipe off any excess.
- o Coat the threads, nuts and washers used in the clamp mechanism with a small amount of graphite spray.
- o Replace the SCR cell carefully, ensuring that its orientation and polarity are correct. Use care when placing the cell on the dowel pin to prevent scratching the cell post surface. Rotate the cell back and forth a few times to ensure proper seating.
- o Reassemble the SCR cell assembly, taking care not to damage the cell with the heatsink. Tighten the clamp nuts only finger tight. Ensure that the number of exposed threads on each stud are about equal.
- o Tighten each nut alternately in accordance with Table I.

TABLE I

SCR Part No.	Number/Turns*	Increments
104X125DA137	1-3/8	1/4 Turn
104X125DA123	3/4	1/4 Turn
104X125DA089	1/2	1/4 Turn

*from finger tight

- o Replace the repaired stack assembly following in reverse the procedure outlined for removal

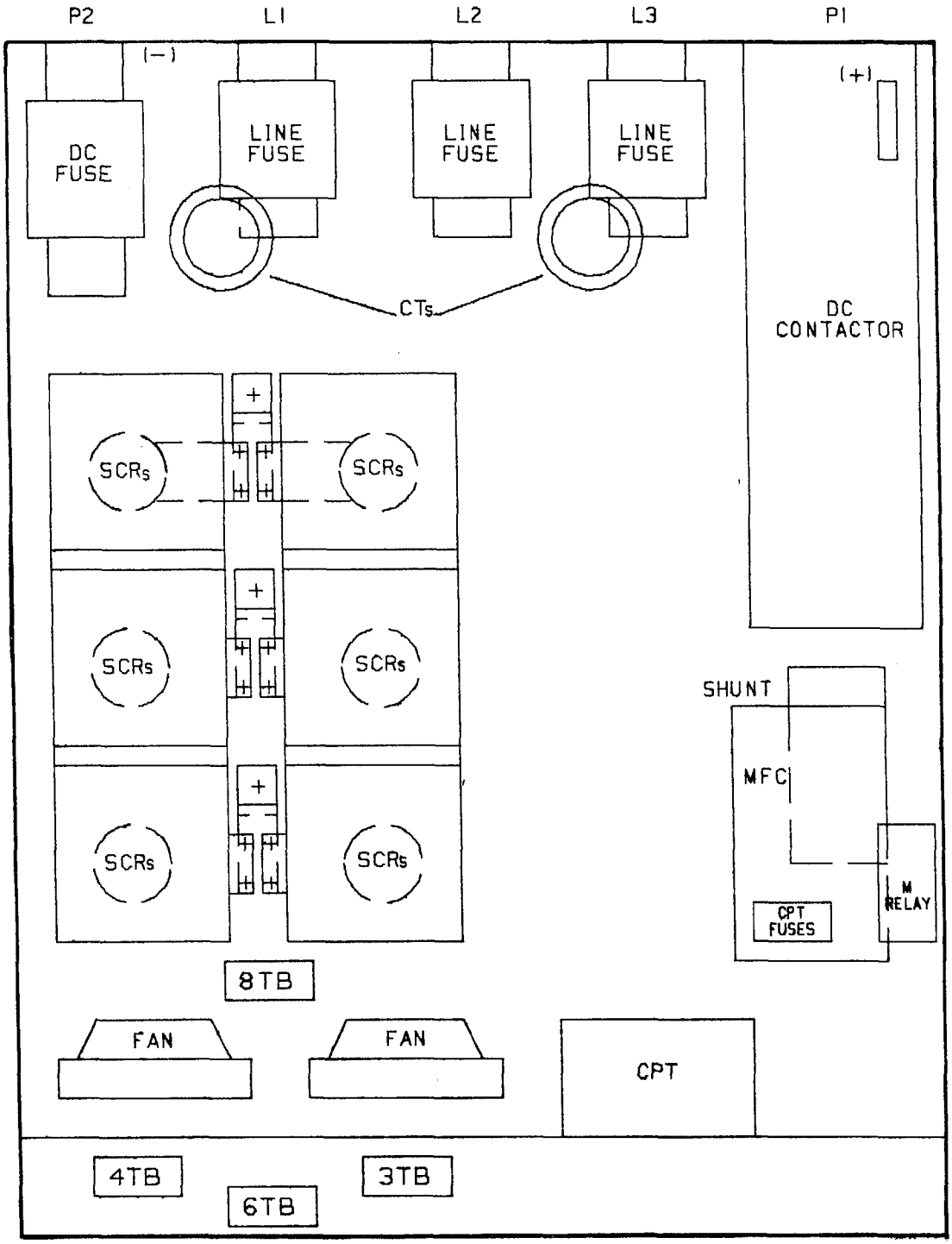
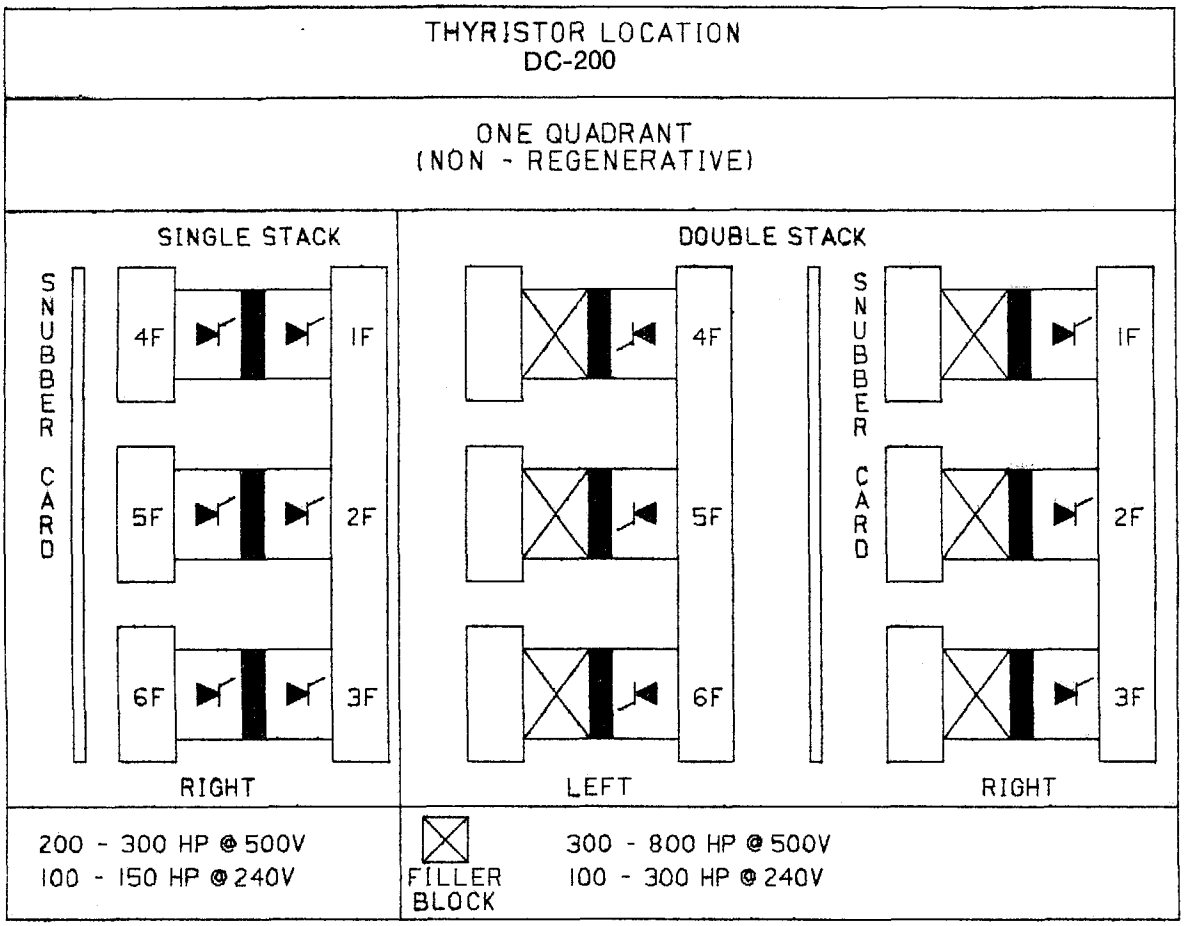


FIGURE 30
SCR LOCATION — GENERAL

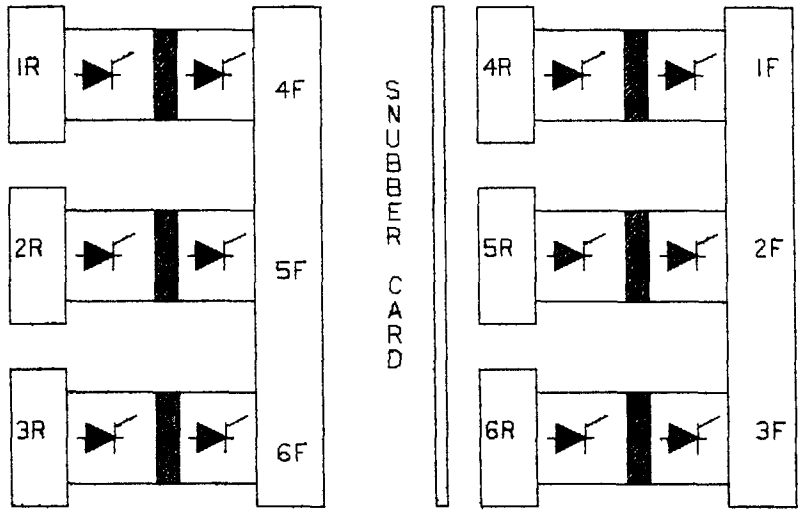


FUSE REPLACEMENT 230VAC					FUSE REPLACEMENT 460VAC				
HP	NAME	FUSE AMPS	FUSE VOLTS	TYPE	HP	NAME	FUSE AMPS	FUSE VOLTS	TYPE*
100 TO 150	FUI-3	500	500	A50P	300	FUI-3	500	500	A50P
					400	FUI-3	800	500	A50P
200	FUI-3	800	500	A50P	500 TO 700	FUI-3	1000	500	A50P
250 TO 300	FUI-3	1000	500	A50P					
					800	FUI-3	1200	500	A50P
* GOULD-SHAWMUT OR BRUSH ELECTRIC OR EQUIVALENT									

FIGURE 31
SCR LOCATION—NON-REGENERATIVE

THYRISTOR LOCATION
DC-200

FOUR QUADRANT (REGENERATIVE)



300 - 800HP AT 500V
100 - 300HP AT 240V

FUSE REPLACEMENT 230VAC					FUSE REPLACEMENT 460VAC				
HP	NAME	FUSE AMPS	FUSE VOLTS	TYPE	HP	NAME	FUSE AMPS	FUSE VOLTS	TYPE✳
100 TO 150	FU1-3	500	500	A50P	300	FU1-3	500	500	A50P
					400	FU1-3	800	500	A50P
200	FU1-3	800	500	A50P	500 TO 700	FU1-3	1000	500	A50P
250 TO 300	FU1-3	1000	500	A50P					
100	FU4	400	500	A50P	800	FU1-3	1200	500	A50P
125	FU4	500	500	A50P	300 TO 400	FU4	700	700	A70P
150	FU4	600	500	A50P					
200	FU4	800	500	A50P	500 TO 700	FU4	700✳✳	700	A70P
250	FU4	1000	500	A50P					
300	FU4	700✳✳	500	A50P	800	FU4	800✳✳	700	A70P

✳ GOULD-SHAWMUT OR BRUSH ELECTRIC OR EQUIVALENT ✳✳ QTY 2 PER DRIVE

FIGURE 32
SCR LOCATION — REGENERATIVE

APPENDIX II
INDEX
DC-200 Modification Kit Instructions

KIT	DESCRIPTION	INSTRUCTIONS
7V _ _ CSI, CLI	Remote Instrument Kit	36A349267AD
7V _ _ CIE	Remote Instrument Enclosure Kit	36A349267AG
7V _ _ CPS	Speed Potentiometer	36A349267AN
7V _ _ CPF	Process Interface Kit	36A349267AP

GENERAL ELECTRIC COMPANY
SPEED VARIATOR PRODUCTS OPERATION
ERIE, PENNSYLVANIA 16531

GENERAL ELECTRIC

GEK-85770 (11/86) 1M (P)